Self-Consistent Proportional Per-Node Damping for Reachability Ranking in Uncertain Graphs: Exact Theory, Dynamical Obstructions, and Hybrid Dual-Operator Polishing

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Abstract

Damped diffusions rank the vertices of an uncertain graph $G = (V, E, p)$ by expected reachable mass $y_i^* = \mathbb{E}\left[\sum_{v \in C(i)} x_v\right]$ at the cost of a few sparse matrix–vector products, but they apply one damping factor to every vertex. Our earlier work calibrated that factor from the mean escape probability [4] and reported that a per-vertex damping vector fitted by an oracle gains a further +0.062 in Spearman ρ (+0.341 on grids). This paper studies a parameter-free way to set per-vertex damping, the Proportional Relative Influence Principle $\alpha_i/(1-\alpha_i) = (y_i - x_i)/x_i$, coupled with the diffusion update $y_i = (1-\alpha_i)x_i + \alpha_i S_i$. The coupled system factorises exactly as $(y_i - x_i)(y_i + x_i - S_i) = 0$ (Theorem 1). We prove six properties of its realizations: finite-time collapse to the prior on substochastic operators, absorption with quadratic contraction for the lagged form, instability of every positive self-consistent solution, a Lehmer-mean contraction whose fixed point falls below the prior, a sign obstruction under which the law lowers well-connected vertices wherever their neighbour signal is below their prior, and degeneracy of the pull variant under uniform priors; an exact log-dispersion identity replaces the prior-uniformity heuristic. A pre-registered evaluation on 688 instances, with every configuration frozen on a development panel and Holm correction over 21 configurations, confirms the theory: against the calibrated scalar rule the anchored law changes ρ by −0.012, the hybrid mode as written by −0.116, and the law evaluated with the true y^* by −0.356. Measured out of sample, the per-node headroom is +0.028. What does gain is the hybrid architecture used as a dual-operator ensemble: blending the calibrated diffusion with a few sweeps of propagation on the raw probability adjacency improves ρ by +0.051 to +0.056 on held-out instances and transfers to real networks and to $n = 900$. Matched controls attribute the gain to the low correlation of the two operators' rank errors rather than to the proportional law. We also correct two claims of [4]: the calibration is prior-dependent, and a batched Monte Carlo sampler is fast enough that the deterministic ensemble is cheaper than equally accurate sampling only beyond a family-dependent graph size.

1 Introduction

Let $G = (V, E)$ be an undirected graph on n vertices in which every edge $(i, j) \in E$ is present independently with probability $p_{ij} \in (0, 1]$, and let $x \in \mathbb{R}^n_{>0}$ with $\sum_i x_i = 1$ be a prior over the vertices. In a sampled world $G' \sim G$ let $C(i)$ be the connected component of vertex i. The *expected reachable mass*

$$y_i^* = \mathbb{E}_{G' \sim G}\Big[\sum_{v \in C(i)} x_v\Big] \tag{1}$$

is the single-seed influence spread of the node-weighted independent cascade model [1] and, under the uniform prior, the edge-percolated component (EPC) centrality [2, 3] divided by n. Computing it exactly is #P-hard, since it contains two-terminal network reliability [6]; in practice it is estimated by Monte Carlo sampling of worlds, by message passing [5], or by cheap deterministic heuristics that only need to *rank* the vertices correctly. Damped diffusions of the personalised PageRank family [8–11] are the most common heuristics of the last kind.

A damped diffusion mixes the prior with a propagated signal, $y \leftarrow (1-\alpha)x + \alpha Ty$, and the damping factor α decides how far mass travels. In [4] we showed that α need not be inherited from web search or tuned against ground truth: the clipped-linear rule

$$\alpha_0 = \mathrm{clip}\big(0.502 + 0.558\,\bar{P}_{\mathrm{esc}},\ 0.4,\ 1\big), \qquad P_{\mathrm{esc}}(i) = 1 - \prod_{j \in N(i)} (1 - p_{ij}), \qquad \bar{P}_{\mathrm{esc}} = \tfrac{1}{n}\textstyle\sum_i P_{\mathrm{esc}}(i), \tag{2}$$

recovers 98.2% of a per-instance scalar oracle. The same study reported that a *per-vertex* damping vector fitted by coordinate ascent beats the best scalar damping by +0.062 in Spearman's ρ overall and by +0.341 on regular grids, where the one-hop escape probability is identical for all interior vertices. If even part of that headroom could be captured without supervision, a diffusion could adapt its damping to every vertex.

The companion manuscripts of this paper proposed such a mechanism, the *Proportional Relative Influence Principle*: at every vertex the odds of propagating rather than retaining mass should equal the vertex's external influence relative to its own prior,

$$\begin{cases} y_i = (1-\alpha_i)\,x_i + \alpha_i\,S_i & \text{(diffusion equation)}, \\ \dfrac{\alpha_i}{1-\alpha_i} = \dfrac{y_i - x_i}{x_i} & \text{(proportionality equation)}, \end{cases} \tag{3}$$

where $S_i = (Ty)_i$ is the neighbour signal of a propagation operator T. The system is solved in closed form, it suggests three iterative realizations, and it can be executed either from scratch (Mode (a)) or as a "polishing" stage after a calibrated base diffusion (Mode (b)).

This paper examines the principle as rigorously as we can, through exact analysis and through a pre-registered experiment, and reports what survives. The result is largely negative, and the negative part is explained: the coupled system is not satisfied by the target (1) on any normalised operator, its realizations either collapse onto the prior or move away from their self-consistent point, and the law raises or lowers a vertex according to the sign of $S_i - x_i$ rather than its reachability. The experiment confirms each of these mechanisms. The architecture of Mode (b), however, turns out to be useful for a reason that the principle did not anticipate: combining two propagation operators whose errors are weakly correlated. Our contributions are:

(i) **Exact algebra and a nested generalisation** (Section 3). We prove the factorisation of the coupled system (Theorem 1), identify what it requires of the operator scale, define the realizations precisely and embed the law in an anchored power family $\mathrm{AP}(\eta)$ that contains the calibrated scalar rule at $\eta = 0$.

(ii) **Dynamical obstructions** (Section 4). Propositions A–F establish finite-time collapse, absorption, instability with a closed-form spectral radius, a Lehmer-mean contraction below the prior, a sign obstruction (with a rigorous version for linear per-node diffusions) and a degeneracy of the pull variant. Proposition $2'$ turns the prior-uniformity hypothesis into an exact identity.

(iii) **A pre-registered evaluation** (Sections 5–6) on 688 instances in five panels, with 2504 configurations searched on a development panel, 21 frozen before any held-out result was seen, paired tests with Holm correction and matched controls that isolate the contribution of the per-node law.

(iv) **A re-measurement of the per-node headroom**: +0.028 out of sample (+0.051 in sample), +0.018 on grids, concentrated in the dense, heterogeneous-prior regime.

(v) **Dual-operator polishing**: a constructive outcome of Mode (b) that recovers 99.6% of the scalar oracle on held-out instances, together with the evidence that its gain comes from operator complementarity.

(vi) **Two corrections to [4]** — the calibration depends on the prior, and batched Monte Carlo is far cheaper than assumed — and a measured, family-dependent cost crossover between deterministic ranking and sampling.

Section 2 reviews related work, Section 3 develops the principle and its algebra, Section 4 the dynamical theory, Section 5 the experimental protocol and Section 6 the results. Section 7 discusses their implications, Section 8 the threats to validity, and Section 9 concludes.

2 Related work

Reachability in uncertain graphs. Uncertain (probabilistic) graphs with independent edges are the standard model for unreliable links and uncertain interactions; queries such as nearest neighbours [7] and connectivity require reasoning over exponentially many worlds, and two-terminal reliability is #P-complete [6]. Expected reachable mass coincides with single-seed influence spread under the independent cascade [1], for which sampling-based estimators dominate influence maximisation [12, 13]. Message passing for percolation on sparse networks [5, 14] gives accurate component statistics on locally tree-like graphs, and the escape probability P_{esc} used in [4] is exactly its correlation-free zeroth iterate. Edge-percolated component centrality [2, 3] is the uniform-prior special case of (1).

The damping factor. PageRank [8], personalised and topic-sensitive PageRank [9, 10] and their many variants [11] share the damping (or restart) parameter α. Its effect on rankings has been studied as a function [15] and as a source of ranking instability [16]; Constantine and Gleich [17] treat α as a random variable and average over it, and mean-field analyses describe personalised PageRank on random graph models [18]. Our earlier calibration rule [4] predicts the best scalar α for reachability ranking from $\bar{P}_{\mathrm{esc}}$.

Node-dependent damping and restart. Allowing the restart probability to depend on the vertex has been studied from several angles. Avrachenkov, van der Hofstad and Sokol [19] introduced two generalisations of personalised PageRank with node-dependent restart, one based on the proportion of visits to each vertex before restart and one on the vertex visited just before restart; the two coincide for a constant restart probability, and both connect direct and reverse personalised PageRank. Partially absorbing random walks [20] assign each vertex its own absorption rate. Supervised restart probabilities have been learned for ranking and link prediction [21], and generalised PageRank methods learn hop-dependent diffusion coefficients for seed expansion and graph neural networks [22–24]; Katz's status index [25] is the undamped ancestor of these truncated walks. In all of these works the per-vertex (or per-hop) parameters are exogenous: they are fixed by design or learned with supervision. The principle studied here makes α_i an *endogenous* function of the score being computed. That feedback produces both the exact algebra of Section 3 and the dynamical obstructions of Section 4.

Combining rankings. Score and rank averaging are the simplest rank-aggregation methods [26]; ensembles help when their members' errors are weakly correlated [27]. Our dual-operator polishing is a two-member ensemble of this kind, and we measure the correlation of its members' errors directly.

3 The principle and its exact algebra

3.1 Operators and the calibrated diffusion

Write $N(i)$ for the neighbours of i and $P_{\mathrm{esc}}(i)$ for its escape probability (2), the probability that at least one of its edges is present. Following [4] we use the row-stochastic matrix W with $w_{ij} \propto -\ln(1-p_{ij})$ on the edges ($\sum_j w_{ij} = 1$ for non-isolated i), the escape-scaled pull operator $EW = \mathrm{diag}(P_{\mathrm{esc}})\,W$, the push operator $W^\top$ of standard personalised PageRank, the raw probability adjacency $P = (p_{ij})$ and its escape-scaled version $EP = \mathrm{diag}(P_{\mathrm{esc}})\,P$. The rows of W sum to one and those of EW to $P_{\mathrm{esc}}(i) \le 1$; the rows of P sum to the expected degree $\sum_j p_{ij}$, which may exceed one.

For a damping vector $\alpha \in [0,1]^n$ and $D = \mathrm{diag}(\alpha)$, the *normalised damped diffusion* is the Perron vector of

$$B(\alpha) = (I - D)\,x\mathbf{1}^\top + DT, \tag{4}$$

that is, the fixed point of $y \leftarrow \mathscr{N}\big((1-\alpha)\circ x + \alpha \circ Ty\big)$ with $\mathscr{N}(v) = v/\mathbf{1}^\top v$. We compute it with the lazy power iteration $y \leftarrow \mathscr{N}\big(\frac{1}{2}(y + B(\alpha)y)\big)$, which has the same fixed point and also converges on bipartite graphs. With $T = EW$ and the scalar α_0 of (2) this is the *calibrated rule* of [4], the reference method of this paper. A score vector is evaluated by Spearman's ρ with (1); tied scores receive average ranks and a constant score vector receives $\rho = 0$.

3.2 The principle and Theorem 1

Theorem 1 (exact solution of the coupled system)**.** *Let $x_i > 0$ and $S_i \geq 0$. A pair (y_i, α_i) with $y_i > 0$ and $\alpha_i \neq 1$ solves* (3) *if and only if $\alpha_i = 1 - x_i/y_i$ and*

$$(y_i - x_i)\,(y_i + x_i - S_i) = 0. \tag{5}$$

The solutions are the trivial branch $(y_i, \alpha_i) = (x_i, 0)$ and, when $S_i > x_i$, the active branch

$$y_i = S_i - x_i, \qquad \alpha_i = \frac{S_i - 2x_i}{S_i - x_i} = 1 - \frac{x_i}{S_i - x_i}. \tag{6}$$

The active branch has $\alpha_i \in (0,1)$ if and only if $S_i > 2x_i$; the two branches coincide at $S_i = 2x_i$, where the discriminant $(S_i - 2x_i)^2$ vanishes.

Proof. Adding one to both sides of the proportionality equation gives $1/(1-\alpha_i) = y_i/x_i$, so $\alpha_i = 1 - x_i/y_i$ (this uses $\alpha_i \neq 1$ and $y_i \neq 0$), and conversely this value of α_i satisfies the proportionality equation. Substituting it into the diffusion equation and multiplying by $y_i > 0$ gives $y_i^2 = x_i^2 + (y_i - x_i)S_i$, that is $(y_i - x_i)(y_i + x_i) - S_i(y_i - x_i) = 0$, which is (5). The root $y_i = x_i$ gives $\alpha_i = 0$. The root $y_i = S_i - x_i$ is positive exactly when $S_i > x_i$ and gives $\alpha_i = 1 - x_i/(S_i - x_i) = (S_i - 2x_i)/(S_i - x_i)$, which lies in $(0,1)$ exactly when $S_i > 2x_i$. Expanding (5) gives $y_i^2 - S_i y_i + x_i S_i - x_i^2 = 0$ with discriminant $S_i^2 - 4x_i S_i + 4x_i^2 = (S_i - 2x_i)^2$. □

Theorem 1 is the algebraic core of the companion manuscripts, and it is correct: in our implementation both branches satisfy both equations to a relative residual of 3.2×10^{-12}. Its interpretation, however, deserves care.

Remark 1 (what the system asks of the operator)**.** On the active branch the score $y_i = S_i - x_i$ *decreases* in the vertex's own prior for a fixed neighbour signal, whereas the target $y_i^* = x_i + \mathbb{E}\big[\sum_{v \in C(i)\setminus\{i\}} x_v\big]$ increases in it. The target satisfies the coupled system on the active branch only if $Ty^* = y^* + x$, that is, only if the operator maps the target to itself plus the prior. Since $y^* + x > y^*$ componentwise, the Collatz–Wielandt bound [28] then forces $\rho(T) > 1$: no substochastic operator, and in particular none of the normalised diffusions used in practice, can satisfy the system at the target. The coupled system is therefore a constraint on the scale of the operator, not a property of reachability.

3.3 Three realizations and their scales

An iterative solver evaluates $S^{(t)} = Ty^{(t)}$ and updates every vertex in parallel. Following the manuscripts we study three realizations, all of the form $y^{(t+1)} = R\big((1 - \alpha^{(t+1)}) \circ x + \alpha^{(t+1)} \circ S^{(t)}\big)$ for a scale map R discussed below:

Realization A (simultaneous root).
$\alpha_i^{(t+1)} = \mathrm{clip}\big((S_i^{(t)} - 2x_i)/(S_i^{(t)} - x_i), 0, \alpha_{\max}\big)$ if $S_i^{(t)} > 2x_i$ and 0 otherwise. Without the cap this is the upper envelope $y_i^{(t+1)} = \max\big(x_i, S_i^{(t)} - x_i\big)$ of Theorem 1.

Realization B (time-lagged).
$\alpha_i^{(t+1)} = 1 - x_i/y_i^{(t)}$, implemented as $(y_i^{(t)} - x_i)_+ / \big((y_i^{(t)} - x_i)_+ + x_i\big)$ so that α is never negative; hence $y_i^{(t+1)} = x_i^2/y_i^{(t)} + (1 - x_i/y_i^{(t)})S_i^{(t)}$ while $y_i^{(t)} \geq x_i$. Its fixed points are exactly the solutions of (5).

Realization C (direct neighbour signal).
$\alpha_i^{(t+1)} = \gamma S_i^{(t)} / \big(x_i + \gamma S_i^{(t)}\big)$; for the manuscripts' $\gamma = 1$ the update is the Lehmer mean $y_i^{(t+1)} = L_2(x_i, S_i^{(t)}) = (x_i^2 + S_i^{(t)2})/(x_i + S_i^{(t)})$ [29].

All realizations start from the one-hop warm start $y^{(0)} = x + Tx$. Because the law compares y_i with x_i, the scale of y matters, and we study four conventions for R: the *raw* scale (R is the identity), the L_1 scale ($R = \mathscr{N}$), and the *excess* scale $\mathrm{ex}(\gamma)$,

$$R(y) = x + \gamma \frac{z}{\frac{1}{n}\sum_j z_j/x_j}, \qquad z = (y - x)_+, \tag{7}$$

which keeps every vertex at or above its prior and fixes the mean excess odds $\frac{1}{n}\sum_j (y_j - x_j)/x_j$ at $\gamma \in \{1, 4, 16\}$. The operators of Section 3.1 give the two variants of the manuscripts: $S = EWy$ (escape-scaled pull) and $S = Wy$ ("standard PPR" as written, i.e. pull without escape scaling); we add the push operator $W^\top$, the raw adjacency P and EP.

3.4 The anchored power family

On a normalised scale the excess $y_i - x_i$ has no fixed origin: half of the vertices fall below their prior after L_1 normalisation, and multiplying y by a constant changes $(y_i - x_i)/x_i$ non-uniformly. The ratio $q_i = y_i/x_i = 1 + (y_i - x_i)/x_i$ does not have this problem, because a global rescaling multiplies all q_i by the same factor. We therefore embed the law in the family

$$\frac{\alpha_i}{1-\alpha_i} = \exp\Big(c + \eta\Big(\log q_i - \frac{1}{n}\sum_j \log q_j\Big)\Big), \qquad q_i = \frac{y_i}{\tilde{x}_i}, \qquad \tilde{x} = (1-\tau)x + \frac{\tau}{n}\mathbf{1}, \tag{8}$$

where c is the unique constant with $\frac{1}{n}\sum_i \alpha_i = \alpha_0$ (the left-hand mean is strictly increasing in c; we solve for it by a safeguarded Newton iteration). We call (8) the *anchored proportional law* $\mathrm{AP}(\eta)$. It has three properties that make it the fairest test of the principle: at $\eta = 0$ it *is* the calibrated scalar rule ($\alpha_i \equiv \alpha_0$); it is invariant to the global scale of y; and at $\eta = 1$ it sets the damping odds proportional to q_i, which is the principle with its global level fixed by the rule of [4] (the additive unit that separates q_i from $(y_i - x_i)/x_i$ is not identifiable on a normalised scale and is absorbed by the anchor). The regulariser τ implements the denominator regularisation proposed in the manuscripts; $\tau = 0$ unless stated. $\mathrm{AP}(\eta)$ is run either to its self-consistent fixed point (joint lazy iteration of y and α) or as a fixed-α polish.

3.5 Execution modes

Mode (a) runs a realization or $\mathrm{AP}(\eta)$ from the warm start for T steps or to convergence. *Mode (b)* first computes a base vector y^{base} — the calibrated rule, or two iterations of the message passing of [5] — embeds it on the excess scale, $y^{(0)} = x + \gamma z$ with $z = \mathscr{N}(y^{\mathrm{base}})/\big(\frac{1}{n}\sum_j \mathscr{N}(y^{\mathrm{base}})_j/x_j\big)$, runs K sweeps of a law on some operator to obtain y^{prop}, and blends

$$\begin{aligned} y^{\mathrm{final}} &= (1-\beta)\,\mathscr{N}\big(y^{\mathrm{base}}\big) + \beta\,\mathscr{N}\big(y^{\mathrm{prop}}\big) && \text{(score blend)}, \\ r^{\mathrm{final}} &= (1-\beta)\,r\big(y^{\mathrm{base}}\big) + \beta\,r\big(y^{\mathrm{prop}}\big) && \text{(rank blend)}, \end{aligned} \tag{9}$$

where $r(\cdot)$ denotes the vector of ranks. The manuscripts' nominal Mode (b) — Realization B on the excess scale, $K = 1$ sweep, $\beta = 0.5$ — is reported as *Mode (b) as written*.

4 Dynamical theory

Throughout this section $T \geq 0$ entrywise, inequalities between vectors hold componentwise, and T is *row-substochastic* if $T\mathbf{1} \leq \mathbf{1}$; W and EW are row-substochastic, while the push operator $W^\top$ has column sums one and spectral radius one. Unless stated otherwise the realizations run on the raw scale with $\alpha_{\max} = 1$. Figure 1 and Table 13 (Appendix A) show the corresponding measurements on the development panel.

4.1 Collapse and absorption on contractive operators

Proposition A (finite-time collapse of Realization A)**.** *Let T be row-substochastic, $x = \frac{1}{n}\mathbf{1}$, and run Realization A from any $y^{(0)} \geq x$. Then $M_t = \max_i y_i^{(t)}$ satisfies $M_{t+1} \leq \max\big(\frac{1}{n}, M_t - \frac{1}{n}\big)$, so $y^{(t)} = x$ for every $t \geq \lceil nM_0 \rceil - 1$. From the warm start $y^{(0)} = x + Tx$ the collapse happens at $t = 1$.*

Proof. Every iterate satisfies $y^{(t)} \geq x$, because the update is $\max(x, S - x)$. Since $T \geq 0$ and $T\mathbf{1} \leq \mathbf{1}$, $S_i = (Ty^{(t)})_i \leq M_t$, hence $y_i^{(t+1)} = \max\big(\frac{1}{n}, S_i - \frac{1}{n}\big) \leq \max\big(\frac{1}{n}, M_t - \frac{1}{n}\big)$. By induction $M_t \leq \max\big(\frac{1}{n}, M_0 - \frac{t}{n}\big)$, which equals $\frac{1}{n}$ once $t \geq nM_0 - 1$; together with $y^{(t)} \geq x$ this gives $y^{(t)} = x$. The trivial point is then fixed, because $Tx \leq x$ implies $S - x \leq 0 < x$. For the warm start, $y^{(0)} = x + Tx \leq 2x$, so $M_0 \leq 2/n$ and one step suffices. □

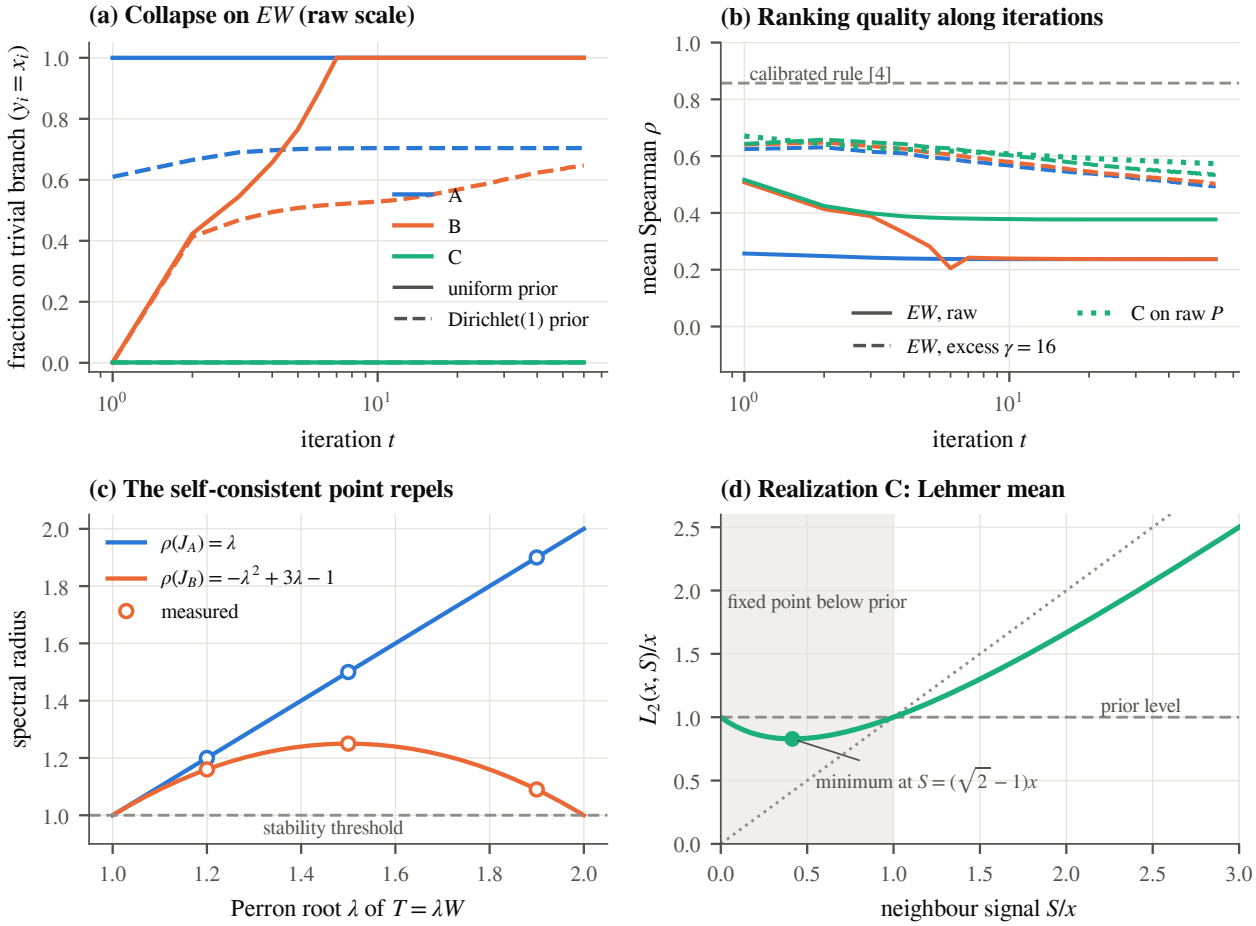


Figure 1: Dynamics of the realizations (development panel). (a) Fraction of vertices on the trivial branch $y_i = x_i$ for Realizations A, B and C on *EW* at the raw scale: under uniform priors A collapses at $t = 1$ (Proposition A) and B is absorbed within a few steps (Proposition B). (b) Mean Spearman ρ along the iterations, colours as in (a); solid: *EW*, raw scale; dashed: *EW*, excess scale $\gamma = 16$; dotted: Realization C on the raw adjacency P. Every configuration stays below the calibrated rule of [4]. (c) Spectral radii of the Jacobians at the positive self-consistent point for $T = \lambda W$; both exceed one on $1 < \lambda < 2$ (Proposition C); circles are numerical values. (d) The Realization C update $S \mapsto L_2(x, S)/x$: it falls below the prior when $S < x$ and decreases on $S < (\sqrt{2} - 1)x$ (Proposition D).

Under uniform priors 100% of the vertices are on the trivial branch after one step on *EW* (Figure 1a), and over all tested instance–operator pairs the largest relative excess left after one step is $1.4{\times}10^{-15}$, i.e. round-off. Under heterogeneous priors a vertex stays active only while $(Ty)_i > 2x_i$, which requires neighbours with much more prior mass than the vertex itself; 30% of the vertices remain active under Dirichlet(1) priors. The condition favours vertices whose own prior is small relative to their neighbours', not vertices that are well connected.

Proposition B (absorption and quadratic contraction of Realization B)**.** *Let T be row-substochastic and run Realization B. (i) If $y_i^{(t)} \le x_i$ for some t, then $y_i^{(s)} = x_i$ for every $s > t$. (ii) If $x = \xi \mathbf{1}$ is uniform and $m_t = \max_i \left(y_i^{(t)} - \xi\right)_+$, then*

$$m_{t+1} \le \frac{m_t^2}{\xi + m_t}, \tag{10}$$

so m_t decreases to zero, quadratically once $m_t \ll \xi$, and $y^{(t)} \to x$.

Proof. (i) If $y_i^{(t)} \le x_i$ the implemented law gives $\alpha_i^{(t+1)} = 0$, hence $y_i^{(t+1)} = x_i$ and, repeating the argument, $y_i^{(s)} = x_i$ for all $s > t$. (ii) Let $e_i = (y_i^{(t)} - \xi)_+ \le m_t$. Then $\alpha_i^{(t+1)} = e_i/(e_i + \xi) \le m_t/(m_t + \xi)$ because $e \mapsto e/(e + \xi)$ is increasing, and $S_i \le \max_j y_j^{(t)} \le \xi + m_t$ because T is row-substochastic. If $S_i \le \xi$ the update $(1 - \alpha_i)\xi + \alpha_i S_i$ does not exceed ξ. Otherwise $y_i^{(t+1)} - \xi = \alpha_i^{(t+1)}(S_i - \xi) \le \frac{m_t}{m_t + \xi} m_t$. This is (10). Since

$m^2/(\xi+m) < m$ for $m > 0$ and $m_t \le m_0$, the excess contracts at least geometrically with ratio $m_0/(\xi+m_0) < 1$, and quadratically when $m_t \ll \xi$; vertices below ξ return to ξ at the next step by (i). □

On *EW* under uniform priors 77% of the vertices are absorbed by $t = 5$ and 100% by $t = 20$. Absorption is what the lagged form was designed to avoid: the manuscripts introduced it precisely because it has no threshold at $2x_i$, but on a contractive operator the excess it feeds on shrinks at every step.

4.2 Instability of the self-consistent solution

Proposition C (positive self-consistent solutions are repelling)**.** *Let $T \ge 0$ and let $y > 0$ be an all-active fixed point, $y = Ty - x$ with $Ty > 2x$.*

(i) $\rho(T) > 1$. *In particular no such point exists when $\rho(T) \le 1$, e.g. for W, EW and $W^\top$.*

(ii) The Jacobians of the Realization A and B maps at y are $J_A = T$ and $J_B = \mathrm{diag}(x/y) + \mathrm{diag}(\mathbf{1} - x/y)\,T$, and $\rho(J_A) > 1$, $\rho(J_B) > 1$; the point is unstable for both realizations.

(iii) For $T = \lambda W$ with W row-stochastic and $x = \xi\mathbf{1}$, the vector $y = x/(\lambda - 1)$ is an all-active fixed point if and only if $1 < \lambda < 2$, and then

$$\rho(J_A) = \lambda, \qquad \rho(J_B) = -\lambda^2 + 3\lambda - 1, \tag{11}$$

both larger than one on $(1,2)$.

Proof. We use the Collatz–Wielandt bound [28, Thm. 8.1.26]: if $A \ge 0$, $v > 0$ and $Av \ge cv$, then $\rho(A) \ge c$. (i) $Ty = y + x$ gives $(Ty)_i/y_i = 1 + x_i/y_i$, whose minimum exceeds one. (ii) On the active set the A map is $y \mapsto Ty - x$, with Jacobian T. The B map is $y_i' = x_i^2/y_i + (1 - x_i/y_i)S_i$ with $S = Ty$; its explicit derivative in y_i is $x_i(S_i - x_i)/y_i^2 = x_i/y_i$ at the fixed point (where $S_i - x_i = y_i$), and its derivative in S_i is $1 - x_i/y_i$, which gives J_B. Both Jacobians are nonnegative, because $y = S - x > x$. Moreover $J_A y = y + x > y$ and

$$J_B y = x + (\mathbf{1} - x/y) \circ (y + x) = y + x \circ \frac{y - x}{y} > y,$$

so $\rho(J_A) > 1$ and $\rho(J_B) > 1$ by the same bound. A fixed point at which the Jacobian has an eigenvalue of modulus greater than one is unstable. (iii) Since $W\mathbf{1} = \mathbf{1}$, $y = c\mathbf{1}$ solves $y = \lambda W y - \xi\mathbf{1}$ if and only if $c = \xi/(\lambda - 1)$, which is positive if and only if $\lambda > 1$ and exceeds ξ (equivalently $S > 2x$) if and only if $\lambda < 2$. Then $x/y = (\lambda - 1)\mathbf{1}$ and $J_B = (\lambda - 1)I + (2 - \lambda)\lambda W$. For $a, b \ge 0$ the spectral radius of $aI + bW$ is $a + b\,\rho(W) = a + b$, which gives (11); finally $-\lambda^2 + 3\lambda - 2 = (\lambda - 1)(2 - \lambda) > 0$ on $(1,2)$. □

At $\lambda = 1.2$, 1.5 and 1.9 the numerically computed spectral radii of J_B are 1.16, 1.25 and 1.09, as (11) predicts (Figure 1c), and a perturbation of relative size 10^{-9} grows until the iterates leave the fixed point. Proposition C and Remark 1 say the same thing from two sides: the active solution needs an expansive operator, and on an expansive operator it repels. Contractive operators therefore collapse (Propositions A and B), and on expansive ones such as *P* the raw-scale iterates either grow along the dominant eigenvector of T or fall onto the trivial branch.

4.3 Realization C is a different law

Proposition D (Realization C contracts to a point below the prior)**.** *Let $\gamma = 1$, let T be row-substochastic, write $f(x_i, s) = (x_i^2 + s^2)/(x_i + s) = L_2(x_i, s)$, and start from any $y^{(0)} > 0$.*

(i) $f(x_i, s) \ge (2\sqrt{2} - 2)\,x_i$ for all $s \ge 0$, with equality if and only if $s = (\sqrt{2} - 1)x_i$; f decreases in s on $[0, (\sqrt{2} - 1)x_i]$ and increases afterwards; and $f(x_i, s) < x_i$ if and only if $0 < s < x_i$.

(ii) For $X \ge \max\left(\max_i x_i, \max_i y_i^{(0)}\right)$ the box $\mathscr{B} = \{y : (2\sqrt{2} - 2)x \le y \le X\mathbf{1}\}$ contains every iterate from $t = 1$ on, and the Realization C map is an ∞-norm contraction on $\mathscr{B}$. The iteration therefore converges to a unique fixed point y^C from every positive start.

(iii) Every vertex with $0 < S_i < x_i$ at y^C has $y_i^C < x_i$, and an isolated vertex ($S_i = 0$, $y_i^C = x_i$) is ranked above any non-isolated vertex with the same prior and $S_i < x_i$.

Proof. (i) With $r = s/x_i$, $f/x_i = (1+r^2)/(1+r)$ has derivative $(r^2+2r-1)/(1+r)^2$, which vanishes at $r = \sqrt{2}-1$, is negative before and positive after; the minimum value is $(4-2\sqrt{2})/\sqrt{2} = 2\sqrt{2}-2$. Also $f < x_i$ if and only if $s^2 < x_i s$. (ii) By (i) every iterate is at least $(2\sqrt{2}-2)x$. The Lehmer mean lies between its arguments, so $y_i^{(t+1)} \le \max(x_i, S_i) \le \max(x_i, \max_j y_j^{(t)}) \le X$, using $S_i \le \max_j y_j^{(t)}$. The map $F(y)_i = f\big(x_i, (Ty)_i\big)$ has Jacobian $\operatorname{diag}(c)\,T$ with $c_i = \partial_s f = 1 - 2x_i^2/(x_i+S_i)^2 \in (-1,1)$ for $S_i > 0$. On $\mathscr{B}$, a vertex with at least one neighbour has $S_i \in \big[(2\sqrt{2}-2)(Tx)_i,\ X\big]$, a compact interval inside $(0,\infty)$, so $|c_i| \le \kappa_i < 1$; an isolated vertex has a zero row. Hence $\|\operatorname{diag}(c)T\|_\infty \le \max_i \kappa_i < 1$ on the convex set $\mathscr{B}$, F is a contraction there by the mean-value inequality, and Banach's theorem gives a unique fixed point and convergence. Boxes for larger X contain those for smaller X, so the fixed point does not depend on X. (iii) follows from (i) applied at the fixed point. □

Three random starts reach the same fixed point to within $8{\times}10^{-17}$ (Figure 1d). For general $\gamma > 0$ the update is $(x_i^2 + \gamma S_i^2)/(x_i + \gamma S_i)$, and part (iii) holds verbatim because $x_i^2 + \gamma S_i^2 < x_i(x_i + \gamma S_i)$ if and only if $S_i < x_i$. Realization C is thus well behaved as a dynamical system, but it is not the proportional law: its odds S_i/x_i compare the *incoming* signal with the prior instead of the accumulated excess, and its fixed point inverts the order of weakly connected and isolated vertices.

4.4 The sign obstruction

Proposition E (sign obstruction)**.** *(i) For any update $u_i = (1-\alpha_i)x_i + \alpha_i S_i$, $\partial u_i/\partial\alpha_i = S_i - x_i$. A law that gives larger α_i to better-connected vertices — as $\alpha_i = 1 - x_i/y_i$ does — raises them only where $S_i > x_i$ and lowers them where $S_i < x_i$. (ii) Let T be row-substochastic, $\alpha \in [0,1)^n$, $D = \operatorname{diag}(\alpha)$, and let $y(\alpha) = (I-DT)^{-1}(I-D)x$ be the linear per-node diffusion with $S = Ty$. Then*

$$\frac{\partial y}{\partial \alpha_k} = (S_k - x_k)\,Ge_k, \qquad G = (I-DT)^{-1} \ge 0, \qquad g_{kk} \ge g_{jk} \ \text{ for all } j, \tag{12}$$

so $\partial(y_k - y_j)/\partial\alpha_k = (S_k - x_k)(g_{kk} - g_{jk})$ has the sign of $S_k - x_k$ or vanishes: raising α_k can move vertex k up the ranking only if $S_k > x_k$. (iii) For $T = EW$ and the uniform prior, $S = EWx$ has $S_i = P_{\mathrm{esc}}(i)/n < x_i$ for every vertex with $P_{\mathrm{esc}}(i) < 1$.

Proof. (i) is immediate. (ii) The rows of DT sum to at most $\max_k \alpha_k < 1$, so $\rho(DT) < 1$ and $G = \sum_{m\ge 0}(DT)^m \ge 0$. Differentiating $y = (I-D)x + DTy$ in α_k gives $(I-DT)\,\partial y/\partial\alpha_k = (S_k - x_k)e_k$. For the column property let $v = Ge_k/g_{kk}$ (note $g_{kk} \ge 1$), so $v_k = 1$ and $v_j = \sum_l (DT)_{jl} v_l$ for $j \ne k$. Suppose $\mu = \max_j v_j > 1$ and let $A = \{j : v_j = \mu\} \not\ni k$. For $j \in A$, $\mu = \sum_l (DT)_{jl} v_l \le \mu \sum_l (DT)_{jl} \le \mu$, so every row of DT indexed by A sums to one and is supported in A; the principal submatrix $(DT)_{AA}$ is then row-stochastic and has spectral radius one, contradicting $\rho\big((DT)_{AA}\big) \le \rho(DT) < 1$. Hence $v \le \mathbf{1}$, i.e. $g_{jk} \le g_{kk}$. (iii) $W\mathbf{1} = \mathbf{1}$ on non-isolated vertices, so $(EW\,\frac{1}{n}\mathbf{1})_i = P_{\mathrm{esc}}(i)/n$. □

Part (iii) means that at the prior itself every vertex sits on the wrong side of the sign condition; at the rule's fixed point on a sparse Erdős–Rényi instance the same holds for 100% of the vertices. For the normalised diffusion of Section 3.1, finite differences agree with the sign rule (the change of a vertex's own score after raising its α has the sign of $S_k - x_k$) on 20 of 20 randomly chosen vertices. Section 6.3 shows that the share of vertices with $S_i > x_i$ predicts, across instances, whether the optimal per-node damping agrees with the law.

4.5 The pull variant without escape scaling

Proposition F (degeneracy of pull diffusion under uniform priors)**.** *Let W be row-stochastic with no isolated vertices, x uniform, and $S = Wy$. Then every iterate of Realizations A, B and C from the warm start $x + Wx$, on the raw, L_1 or excess scale, is a multiple of x: all vertices tie and $\rho = 0$.*

Proof. $Wx = x$. If $y^{(t)} = cx$ then $S = cx$, and the three updates give $\max(1, c-1)\,x$, $\big((1-a)+ac\big)x$ with $a = (1-1/c)_+$, and $\frac{1+\gamma c^2}{1+\gamma c}\,x$; the three scale maps send multiples of x to multiples of x. The warm start is $2x$. □

On the development panel 85% of the standalone runs with $S = Wy$ under uniform priors score exactly $\rho = 0$, and the panel mean is 0.012. The meaningful "standard PPR" variant is the push operator $W^\top$, which [4] used and which we include throughout.

4.6 Proposition 2 re-examined

The companion manuscripts state, as their Proposition 2, that (1) under the uniform prior $\alpha_i = 1 - 1/(ny_i)$ is a strictly monotone, shift-invariant function of the reachability y_i alone, and (2) as the prior localises, $|\partial_{x_i}((y_i - x_i)/x_i)| = y_i/x_i^2 \to \infty$ and $\alpha_i \to 1$ for every non-seed vertex. Part (1) is correct about monotonicity but does not imply a ranking advantage, because by Proposition E a monotone law still acts with the sign of $S_i - x_i$; and the map is not shift-invariant (adding a constant to y changes the α_i non-uniformly). Part (2) is a correct derivative, and its consequence — saturation — is what we observe under seed priors. Under smooth heterogeneous priors the failure is of a different kind, which the following identity makes exact.

Proposition 2′ (log-dispersion identity)**.** *Let $z_i = y_i - x_i > 0$ be the external mass and $r_i = z_i/x_i$ the proportional ratio. Over the vertices let* $\log z$ *and* $\log x$ *have standard deviations $\sigma_z, \sigma_x > 0$ and correlation ϕ, and assume* $\log r$ *is not constant. Then*

$$\operatorname{corr}(\log r, \log z) = \frac{\sigma_z - \phi\,\sigma_x}{\sqrt{\sigma_z^2 + \sigma_x^2 - 2\phi\,\sigma_z\sigma_x}}, \qquad \operatorname{corr}(\log r, -\log x) = \frac{\sigma_x - \phi\,\sigma_z}{\sqrt{\sigma_z^2 + \sigma_x^2 - 2\phi\,\sigma_z\sigma_x}}. \tag{13}$$

As $\sigma_x/\sigma_z \to \infty$ the first tends to $-\phi$ and the second to one: when the log-dispersion of the prior dominates that of the external mass, the ratio orders the vertices by $1/x_i$ and retains no information about z beyond its correlation with the prior.

Proof. $\log r = \log z - \log x$, so

$$\operatorname{Var}(\log r) = \sigma_z^2 + \sigma_x^2 - 2\phi\sigma_z\sigma_x, \quad \operatorname{Cov}(\log r, \log z) = \sigma_z^2 - \phi\sigma_z\sigma_x, \quad \operatorname{Cov}(\log r, -\log x) = \sigma_x^2 - \phi\sigma_z\sigma_x.$$

Dividing gives (13); dividing numerators and denominators by σ_x gives the limits. □

Evaluated on the ground truth of the prior panel (Section 6.4), the rank correlation of r with $1/x$ rises from 0.79 under Dirichlet(10) to 0.98 under Dirichlet(0.3) ($\log_{10}\kappa \approx 8.5$), while its correlation with y^* falls from 0.30 to –0.16. Under seed priors the non-seed majority shares a single prior value, so r stays ordinal with y^* (0.84 and 0.96 for five and one seeds) but saturates: 61% of the vertices receive $\alpha_i > 0.99$ under a single seed, which removes the restart for most of the graph. The denominator regularisation $\tilde{x} = (1-\tau)x + \tau/n$ addresses both failures, and at $\tau = 1$ it removes the prior from the denominator altogether.

5 Experimental protocol

5.1 Benchmark panels

Table 1 lists the five panels (688 instances). Synthetic instances come from 9 families: Erdős–Rényi [30] (mean degree 5), Barabási–Albert [31] ($m = 2$), Watts–Strogatz [32] ($k = 6$, rewiring probability 0.1), square grids and 1:4 rectangular grids (the lattices on which [4] reported its largest headroom), four-block stochastic block models [33], core–periphery graphs [34], uniform random trees and barbells. Edge probabilities are drawn uniformly from one of five ranges, $R_1 = [0.01, 0.06]$, $R_2 = [0.06, 0.18]$, $R_3 = [0.18, 0.40]$, $R_4 = [0.40, 0.70]$ and $R_5 = [0.70, 0.95]$; on Erdős–Rényi graphs alone $\bar{P}_{\text{esc}}$ then spans 0.15 to 0.98.

The development and test panels cross the nine families, the five ranges and two priors (uniform and Dirichlet(1)) with two independent replicates each, at $n = 100$; the scale panel repeats the design at $n = 900$ with one

replicate. The prior panel keeps the graph and the edge probabilities fixed and varies only the prior, over ten types: uniform; Dirichlet(θ) with $\theta \in \{10, 3, 1, 0.3\}$; log-normal with $\sigma \in \{0.5, 1, 2\}$; and softened seed priors in which 5 or 1 seeds share $1-\varepsilon$ of the mass and $\varepsilon = 0.1$ is spread uniformly. It covers five families and the ranges R_2–R_4. The real panel uses Zachary's karate club [35], the Les Misérables co-appearance network [36] and the Davis southern-women network [37], together with two 500-vertex snowball samples of each of Cora, CiteSeer, Amazon-Photo and MS-Academic-CS from the benchmark collection of [38]. Each real graph receives four probability schemes — uniform on $[0.05, 0.2]$, $[0.2, 0.5]$ and $[0.5, 0.9]$, and a symmetrised weighted cascade $p_{ij} = 1/\max(\deg i, \deg j)$ [1] — and both priors.

Table 1: The five benchmark panels. Every instance carries two independent Monte-Carlo ground-truth samples (2,000 worlds each).

Panel	n	Instances	Families	Priors	Role
Development (dev)	100	180	9	uniform, Dir(1)	configuration search
Test	100	180	9	uniform, Dir(1)	confirmatory evaluation
Priors	144–150	150	5	10 types	Proposition 2
Real networks	15–500	88	7	uniform, Dir(1)	transfer (genre)
Scale	900	90	9	uniform, Dir(1)	transfer (size)

5.2 Ground truth

Every instance carries two independent Monte Carlo estimates of (1), y_A^* and y_B^*, each from 2,000 sampled worlds. Worlds are sampled in batches that are stacked into one block-diagonal graph, so that a single connected-components call labels all the worlds of a batch. On small graphs the estimator agrees with exact enumeration of all $2^{|E|}$ worlds within sampling error (largest $|z|$-score 2.6). Every method is scored against y_A^*; every oracle and every "true-y^*" experiment uses y_B^* only, so no fitted quantity sees the evaluation sample. The rank correlation between y_B^* and y_A^*, the *split-sample reference*, measures the reliability of the ground truth. Because both samples are noisy, an exact predictor of y^* would score roughly the square root of this reference, so the reference is not a hard ceiling.

5.3 Methods

References. The calibrated rule (normalised diffusion on EW with α_0) [4]; the same diffusion with the canonical $\alpha = 0.85$; the calibrated push personalised PageRank of [4] ($W^\top$ with $\alpha = \mathrm{clip}(0.241 + 0.859\,\bar{P}_{\mathrm{esc}}, 0.4, 1)$); the naive local damping $\alpha_i = P_{\mathrm{esc}}(i)$; and two iterations of the message passing of Karrer, Newman and Zdeborová [5].

Oracles. The scalar oracle picks the best of the 40 values $\alpha \in \{0.025, 0.05, \ldots, 1\}$ for y_B^*. The per-node oracle runs coordinate ascent over $\alpha \in [0,1]^n$ from the scalar oracle's value, with a 14-value grid per coordinate, at most three sweeps and a time cap (120 s, or 240 s on the 40 grid, rectangular-grid, Erdős–Rényi and Barabási–Albert instances of the scale panel on which it is run). Because it is fitted on y_B^* and scored on y_A^*, it estimates the out-of-sample headroom of per-node damping; in sample it is a lower bound on the optimum.

The law with the true y^*. The diffusion on EW with $\alpha_i = 1 - x_i/y_{B,i}^*$, and the anchored law (8) with $q = y_B^*/x$. These remove every source of estimation error from the principle.

5.4 Selection, freezing and statistics

All design choices of the principle were searched on the development panel only: the realization; the operator (EW, W, $W^\top$, P, EP); the scale (raw, L_1, excess with $\gamma \in \{1, 4, 16\}$); the number of steps; $\eta \in \{-1, -0.5, -0.25, 0, 0.25, 0.5, 1, 2\}$; the base (calibrated rule or message passing); $K \in \{1, 2, 5, \infty\}$; $\beta \in \{0.25, 0.5, 0.75, 1\}$; and the blend type. The search covered 760 standalone, 40 anchored, 840 anchored-polishing and 864 realization-polishing configurations, 2504 in all. The winners, the manuscripts' nominal settings and the matched controls described below were frozen as 21 configurations (Table 12, Appendix A) in

a file whose SHA-256 digest begins with `b55fe19a10d3e5de` (2026-10-25 22:41), before any test, prior, real or scale result was examined. Each frozen configuration was then evaluated once on every held-out panel.

Matched controls. To isolate the contribution of the per-node law inside a pipeline, every frozen configuration that runs a realization is re-run with its α vector replaced at every step by its mean (the *mean-field twin*) or by a random permutation of itself; operator, embedding, K and blend stay fixed. Four of the frozen configurations are controls of this kind for the best realization polisher: its mean-field twin, its permutation, and the same pipeline with a constant $\alpha \in \{0.5, 0.9\}$.

Statistics. For each method we report the panel mean of ρ and the paired difference to the calibrated rule, with a 95% percentile-bootstrap interval from 20,000 resamples [39]; wins, ties and losses (W/T/L, with $|\Delta\rho| \leq 10^{-9}$ counted as a tie); and a two-sided Wilcoxon signed-rank test [40] on the untied pairs. Within each panel the p-values of the 21 frozen configurations are Holm-corrected [41]; references and oracles are reported uncorrected. The Wilcoxon test concerns the location of the paired differences and the bootstrap interval their mean; the two can disagree when a few large differences dominate the mean, and we report both. Differences are printed with three decimals, and a signed zero such as -0.000 denotes a difference below 5×10^{-4} in magnitude. Secondary metrics are Kendall's τ-b [42], the overlap of the top 10% and NDCG@10% [43] with relevance y_A^*.

Implementation. All methods are implemented in Python with NumPy and SciPy sparse matrices [44, 45], and synthetic graphs are generated with NetworkX [46]. Every random draw is seeded from a SHA-256 digest of its role, and all timings are single-process wall-clock medians on one machine.

6 Results

6.1 Replication of [4] and the per-node headroom

Table 2 reproduces the calibration of [4] on the development and test panels. Under Dirichlet priors the rule recovers 96.2% of the in-sample scalar oracle and beats the best constant chosen with hindsight (0.851 against 0.684 at $\alpha = 0.900$), as reported there. Under uniform priors it recovers only 91.2%: the oracle's mean α is 0.56 against the rule's 0.84, the single constant $\alpha = 0.600$ scores 0.894 against the rule's 0.851, and even the canonical $\alpha = 0.85$ scores 0.882. The calibration of [4] is therefore prior-dependent: it was fitted where the best damping is high, and it over-damps when the prior is uniform.

Table 2: Replication of the rule of [4] on the development and test panels ($n = 100$). "Best constant": the single α maximising the panel-mean ρ (chosen with hindsight); "Oracle": best α per instance.

Prior	n	Rule	$\alpha = 0.85$	Best constant	Its α	Oracle	% oracle	Rule $\bar{\alpha}$	Oracle $\bar{\alpha}$
all	360	0.851	0.780	0.780	0.875	0.909	93.6	0.84	0.71
Dirichlet(1)	180	0.851	0.678	0.684	0.900	0.884	96.2	0.84	0.85
uniform	180	0.851	0.882	0.894	0.600	0.933	91.2	0.84	0.56

Table 3 re-measures the headroom of per-node damping with the oracle fitted on y_B^* and scored on y_A^*. Out of sample, the per-node oracle exceeds the scalar oracle by +0.028 (in sample by +0.051, so 46% of the apparent gain is overfitting to the fitting sample), by +0.018 on square grids, +0.036 on rectangular grids and +0.069 on trees. The headroom concentrates in the dense, heterogeneous-prior corner (range R_5 with Dirichlet priors: +0.133 averaged over families), where no scalar α ranks well; a gain of the size reported in [4] appears only on individual dense lattice instances (largest +0.347). Per-node damping does have headroom, but it is smaller and more localised than the motivation of the principle assumed.

6.2 Confirmatory evaluation

Table 4 reports every frozen configuration on the held-out test panel together with the references and oracles, and Figure 4a summarises the main methods on all four held-out panels.

Mode (a). Every standalone realization loses to the rule: the best A, B and C configurations change ρ by -0.089, -0.082 and -0.078 (Holm $p = 2.8\times10^{-4}$, 0.002 and 7.7×10^{-4}). The two prior-specific picks do well

Table 3: Per-node damping headroom: a coordinate-ascent oracle fitted on y_B^* and scored on the independent y_A^* (development and test panels).

Family	Scalar oracle	Per-node oracle	Split-sample ref.	Gain (out of sample)	Gain (in sample)
BA	0.949	0.973	0.988	+0.024	+0.032
barbell	0.926	0.949	0.974	+0.023	+0.048
core–periphery	0.961	0.979	0.990	+0.018	+0.024
ER	0.916	0.939	0.968	+0.023	+0.045
grid	0.933	0.950	0.960	+0.018	+0.040
rectangular grid	0.919	0.955	0.970	+0.036	+0.056
SBM	0.870	0.898	0.947	+0.028	+0.057
tree	0.920	0.989	0.994	+0.069	+0.072
WS	0.773	0.786	0.821	+0.013	+0.086
all	0.907	0.935	0.957	+0.028	+0.051

only under the prior they were chosen for. Realization C on the raw adjacency, the best pick for Dirichlet priors, scores 0.872 under Dirichlet priors but 0.395 under uniform ones; Realization B on the push operator, the best pick for uniform priors, scores 0.876 under uniform priors but 0.369 under Dirichlet ones.

The anchored law. AP($\eta = 1$) changes ρ by –0.012 [–0.019, –0.005] (Holm $p = 3.4\times10^{-6}$) and AP($\eta = 0.25$) by –0.002 (Holm $p = 0.006$). Figure 2c shows the whole η-curve. Under Dirichlet priors it peaks at $\eta = 0$, which is the scalar rule itself, on both the development and the test panel, and it declines on the test panel from 0.846 at $\eta = 0$ to 0.818 at $\eta = 1$ and 0.804 at $\eta = 2$; under uniform priors it is flat (range 0.0039 over $\eta \in [-1, 2]$). Per-node variation in the direction of the law never helps.

Mode (b) with the law. Mode (b) as written changes ρ by –0.116 (Holm $p = 2.9\times10^{-8}$): one sweep of Realization B on the excess scale damages the base ranking, and the 50% blend does not repair it. Its anchored counterpart (AP($\eta = 1$), $K = 1$, $\beta = 0.5$) changes ρ by –0.005 (Holm $p = 5.0\times10^{-6}$) and the self-consistent, rank-blended version by –0.002. The best anchored polish found on the development panel uses $\eta = -0.25$, i.e. it points *against* the law, and it is indistinguishable from the rule (+0.0002; ties on most instances, W/T/L 29/119/32).

The law with the true y^*. Setting $\alpha_i = 1 - x_i/y_i^*$ with the independent sample y_B^* removes estimation error altogether. It scores 0.488 (–0.356), and the anchored law with the true y^* scores 0.817 (–0.027). Knowing y^* does not make the law useful: the obstruction lies in the law, not in its inputs.

Mode (b) with a second operator. The frozen configurations that do improve on the rule all combine the calibrated diffusion with a *different* propagation operator. Realization C on the raw adjacency as a polisher ($\gamma = 4$, $K = 5$, $\beta = 0.5$) gains +0.051 [+0.033, +0.070] (Holm $p = 0.001$). Its matched controls gain as much or more: the mean-field twin +0.053, a constant $\alpha = 0.5$ on P +0.052, and a constant $\alpha = 0.9$ on P with a rank blend +0.056 [+0.043, +0.071] (Holm $p = 4.7\times10^{-15}$, W/T/L 135/0/45). The last reaches $\bar{\rho} = 0.901$, 99.6% of the scalar oracle; it closes 93.7% of the gap between the rule and the scalar oracle and 60.5% of the gap to the per-node oracle. Polishing the two-iteration message-passing base with the scalar rule ($\eta = 0$) gains +0.050, while the law in the same pipeline ($\eta = 1$) gains less (+0.042). The permutation control gains +0.027 on average but loses on most instances (W/T/L 77/1/102; Holm $p = 1.000$), an example of the mean and the Wilcoxon location disagreeing.

6.3 The sign obstruction in data

Proposition E predicts that the law can only agree with good per-node damping on vertices with $S_i > x_i$. Figure 2a tests this across the 360 development and test instances. The fraction of vertices with $S_i > x_i$ at the rule's fixed point predicts the rank correlation between the per-node oracle's α^* and the law (Spearman 0.61, $p = 1.9\times10^{-38}$). In the 85 instances where fewer than 10% of the vertices satisfy $S_i > x_i$ the correlation is –0.34 — the optimal damping moves *against* the law — and the law with the true y^* loses 1.21 in ρ, i.e. it inverts the ranking (Figure 2b). Where most vertices satisfy the sign condition the correlation turns positive, but it never exceeds 0.11.

Table 4: Confirmatory evaluation on the held-out test panel ($n = 100$, 180 instances). Every configuration was frozen on the development panel before any test result was examined.

Method	$\bar{\rho}$	Dir.	Unif.	$\Delta\rho$ vs rule [95% CI]	W/T/L	p_{Holm}	Gap %
References and oracles							
Calibrated scalar rule [4]	0.844	0.842	0.846	—	—	—	0
Fixed $\alpha = 0.85$	0.781	0.681	0.881	−0.064 [−0.093, −0.035]	68/0/112	$7.3\times10^{-5\dagger}$	−106
Calibrated push PPR [4]	0.811	0.838	0.785	−0.033 [−0.066, +0.001]	56/0/124	$0.005^{\dagger}$	−55
Local $\alpha_i = P_{\text{esc}}(i)$	0.441	0.690	0.193	−0.403 [−0.503, −0.307]	41/1/138	$1.0\times10^{-18\dagger}$	−670
KNZ message passing (2 it.)	0.754	0.630	0.878	−0.090 [−0.145, −0.037]	85/0/95	$0.074^{\dagger}$	−150
Scalar oracle	0.904	0.877	0.932	+0.060 [+0.044, +0.078]	129/27/24	$5.6\times10^{-21\dagger}$	100
Per-node oracle	0.937	0.931	0.944	+0.093 [+0.071, +0.117]	136/1/43	$1.9\times10^{-18\dagger}$	155
Split-sample reference	0.958	0.964	0.952	+0.114 [+0.089, +0.141]	135/0/45	$9.7\times10^{-19\dagger}$	189
Law with true y^*: $\alpha_i = 1 - x_i/y_i^*$	0.488	0.707	0.269	−0.356 [−0.452, −0.265]	43/0/137	$1.3\times10^{-17\dagger}$	−592
Anchored law with true y^* ($\eta = 1$)	0.817	0.801	0.834	−0.027 [−0.035, −0.019]	18/49/113	$4.1\times10^{-20\dagger}$	−45
The proportional law (frozen on dev)							
A: raw P, excess $\gamma = 16$, $T = 2$	0.755	0.647	0.864	−0.089 [−0.129, −0.049]	60/0/120	2.8×10^{-4}	−148
B: EP, excess $\gamma = 16$, $T = 2$	0.762	0.717	0.807	−0.082 [−0.123, −0.041]	68/0/112	0.002	−136
C: EP, excess $\gamma = 16$, $T = 2$	0.766	0.717	0.815	−0.078 [−0.117, −0.039]	64/0/116	7.7×10^{-4}	−130
C: raw P, $T = 3$ (Dirichlet pick)	0.634	0.872	0.395	−0.211 [−0.311, −0.116]	75/0/105	0.169	−350
B: push, excess $\gamma = 16$, $T = 50$ (uniform pick)	0.623	0.369	0.876	−0.222 [−0.278, −0.165]	43/0/137	4.4×10^{-11}	−368
AP($\eta = 0.25$), fixed point	0.843	0.839	0.846	−0.002 [−0.005, +0.003]	55/48/77	0.006	−3
AP($\eta = 1$), fixed point	0.832	0.818	0.846	−0.012 [−0.019, −0.005]	54/43/83	3.4×10^{-6}	−20
Rule + AP($\eta = -0.25$), $K = 1$, $\beta = 0.25$	0.844	0.843	0.846	+0.000 [+0.000, +0.000]	29/119/32	1.000	0
Rule + AP($\eta = 1$), $K = 1$, $\beta = 0.5$	0.840	0.833	0.846	−0.005 [−0.007, −0.002]	14/113/53	5.0×10^{-6}	−8
Rule + AP($\eta = 1$) s.c., $K = \infty$, $\beta = 0.25$ (rank)	0.843	0.839	0.846	−0.002 [−0.003, −0.001]	49/65/66	0.025	−3
Mode (b) as written: rule + B, $K = 1$, $\beta = 0.5$	0.728	0.602	0.854	−0.116 [−0.147, −0.087]	50/21/109	2.9×10^{-8}	−193
Mode (b) polishers and matched controls							
MP + AP($\eta = 0.25$), $K = 5$, $\beta = 0.75$ (rank)	0.894	0.884	0.903	+0.049 [+0.032, +0.069]	103/0/77	0.001	82
MP + scalar rule, $K = 5$, $\beta = 0.75$ (rank)	0.894	0.887	0.902	+0.050 [+0.033, +0.070]	103/0/77	5.3×10^{-4}	83
MP + AP($\eta = 1$), $K = 5$, $\beta = 0.75$ (rank)	0.887	0.875	0.899	+0.042 [+0.024, +0.062]	99/0/81	0.055	71
Rule + C on raw P, $\gamma = 4$, $K = 5$, $\beta = 0.5$	0.895	0.894	0.897	+0.051 [+0.033, +0.070]	99/3/78	0.001	85
Rule + A on raw P, $\gamma = 1$, $K = 1$, $\beta = 0.5$	0.851	0.783	0.919	+0.007 [−0.014, +0.028]	63/24/93	0.672	11
Rule + C on raw P, $\gamma = 1$, $K = 5$, $\beta = 0.5$ (rank)	0.775	0.899	0.652	−0.069 [−0.120, −0.020]	96/0/84	0.592	−114
Control: C law → per-step mean	0.898	0.895	0.900	+0.053 [+0.036, +0.073]	112/3/65	5.0×10^{-6}	89
Control: C law → permuted	0.871	0.882	0.861	+0.027 [+0.006, +0.050]	77/1/102	1.000	45
Control: constant $\alpha = 0.5$ on raw P	0.896	0.891	0.901	+0.052 [+0.034, +0.071]	112/3/65	4.8×10^{-6}	86
Control: constant $\alpha = 0.9$ on raw P (rank)	0.901	0.903	0.898	+0.056 [+0.043, +0.071]	135/0/45	4.7×10^{-15}	94

$\bar{\rho}$: mean Spearman correlation; Dir./Unif.: by prior type; $\Delta\rho$: paired mean difference to the calibrated rule with a 95% bootstrap confidence interval; W/T/L: wins/ties/losses; p_{Holm}: Wilcoxon signed-rank test, Holm-corrected over the 21 frozen configurations (†: uncorrected p for references); Gap: share of the rule-to-scalar-oracle gap closed; (rank): rank blend; s.c.: self-consistent.

6.4 Prior uniformity

Table 5 and Figure 3 report the prior panel, in which only the prior changes. The difference between AP($\eta = 1$) and the rule is +0.0003 under the uniform prior and falls with dispersion, monotonically within each prior family, to −0.007 under Dirichlet(10), −0.056 under Dirichlet(0.3), −0.057 under log-normal(2) and −0.098 under a single seed; its Spearman correlation with $\log_{10}\kappa(x)$ is −0.31 ($p = 1.2\times10^{-4}$). The direction of the manuscripts' Proposition 2 is confirmed, but there is no regime of advantage: "best under uniform priors" means "harmless under uniform priors". Figure 3b shows the mechanism identified by Proposition 2′, and Figure 3c the effect of regularisation: $\tau = 0.5$ shrinks the loss (e.g. −0.017 under log-normal(2)) and $\tau = 1$ removes it (between −0.004 and +0.005 over all prior types). At $\tau = 1$ the denominator no longer depends on x, and the law reduces to a monotone transform of y anchored at the rule. The dual-operator configurations gain under every prior type (constant $\alpha = 0.9$ on P: +0.033 to +0.052; Realization C on P: +0.007 to +0.049).

6.5 Attribution: the law against its mean-field twin

Table 6 and Figure 4b compare every frozen configuration that runs a realization with its mean-field twin and its permutation. On the test panel the best polisher is slightly *worse* than its twin (−0.0026 [−0.0047, −0.0006]; Holm $p = 1.1\times10^{-6}$), and at $n = 900$ the two are equal (−0.0005 [−0.0043, +0.0043]). On the real panel the

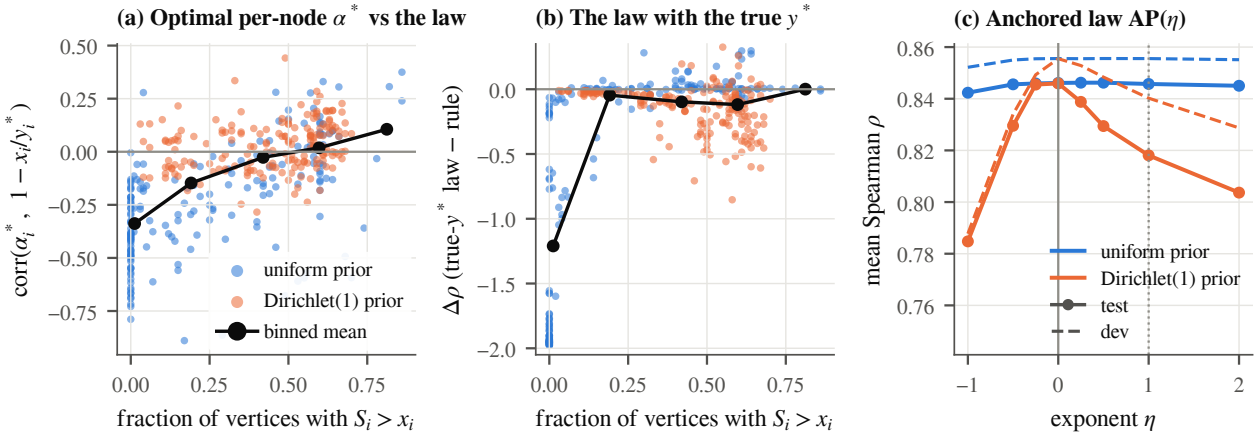


Figure 2: The sign obstruction in data. (a) Rank correlation between the per-node oracle's damping α^* and the law $1 - x_i/y_i^*$, against the fraction of vertices whose neighbour signal exceeds their prior at the rule's fixed point (one point per development or test instance; black: binned means). (b) Difference in ρ between the law evaluated with the true y^* and the calibrated rule. (c) Mean ρ of the anchored law AP(η) on the development (dashed) and test (solid) panels; $\eta = 0$ (solid vertical line) is the scalar rule and $\eta = 1$ (dotted line) the proportional law.

Table 5: The prior-uniformity panel (150 instances; the same graph and edge probabilities across prior types). "Rule" is the mean ρ; the other columns are paired differences $\Delta\rho$ to the rule.

Prior	$\log_{10}\kappa$	Rule	AP τ=0	AP τ=0.5	AP τ=1	Polish (b) τ=0	C on P	α=0.9 on P
uniform	0.0	0.869	+0.000	+0.000	+0.000	+0.000	+0.039	+0.040
Dirichlet(10)	0.8	0.864	−0.007	−0.001	+0.002	−0.003	+0.032	+0.043
log-normal(0.5)	1.2	0.860	−0.017	−0.005	+0.003	−0.007	+0.036	+0.044
Dirichlet(3)	1.5	0.854	−0.013	−0.003	+0.003	−0.006	+0.038	+0.047
log-normal(1)	2.3	0.866	−0.031	−0.004	−0.004	−0.011	+0.049	+0.051
Dirichlet(1)	3.4	0.873	−0.031	−0.006	+0.005	−0.011	+0.038	+0.046
log-normal(2)	4.6	0.844	−0.057	−0.017	−0.004	−0.013	+0.032	+0.049
Dirichlet(0.3)	8.5	0.852	−0.056	−0.017	+0.003	−0.021	+0.039	+0.052
5 seeds, $\varepsilon = 0.1$	2.4	0.867	−0.079	−0.040	+0.005	−0.000	+0.007	+0.033
1 seed, $\varepsilon = 0.1$	3.1	0.818	−0.098	−0.072	−0.003	+0.001	+0.029	+0.042

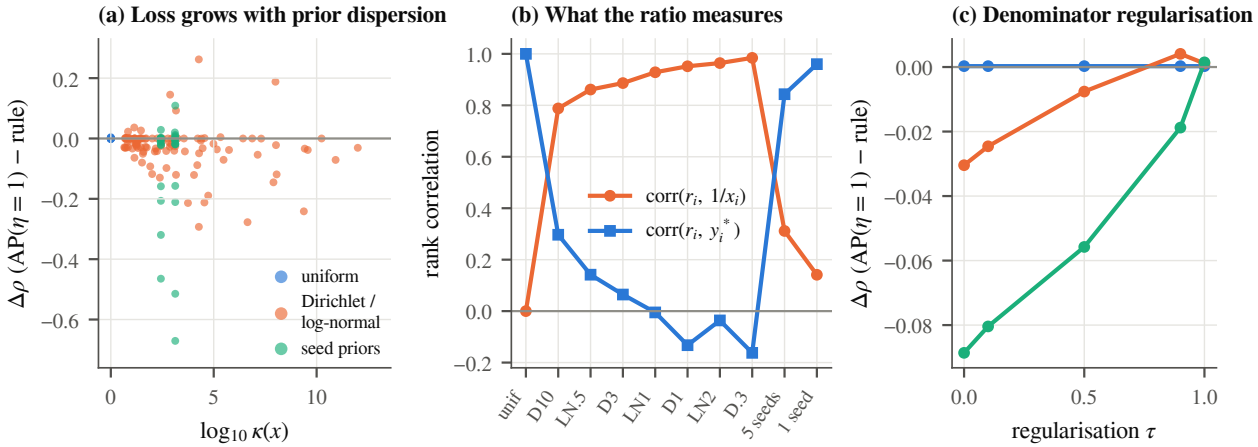


Figure 3: Prior uniformity (prior panel). (a) Paired difference AP($\eta = 1$) minus rule against the dynamic range $\log_{10}\kappa(x)$ of the prior, one point per instance. (b) Mean rank correlation of the ratio $r_i = (y_i^* - x_i)/x_i$ with $1/x_i$ and with y_i^* per prior type, in order of increasing dispersion (D: Dirichlet, LN: log-normal). (c) Effect of the regularisation $\tilde{x} = (1 - \tau)x + \tau/n$ on AP($\eta = 1$) minus rule for uniform, smooth heterogeneous and seed priors.

law retains a small positive effect inside the best polisher (+0.0066 [+0.0031, +0.0108]; Holm $p = 0.032$) and inside the uniform-prior polisher (+0.0075; Holm $p = 0.003$); these are the only positive effects of the law that survive correction, and in both cases the pipeline with a constant $\alpha = 0.9$ is better still (Table 7). Standalone

laws lose to their own twins, often heavily: Realization C on the raw adjacency loses −0.2544 on the test panel, −0.1054 on real networks and −0.2348 at $n = 900$, because replacing its per-node damping by the mean turns it into a plain damped propagation on P. The permuted controls are below the mean-field twins in every row: the assignment of damping values to vertices matters, but the law's assignment beats a uniform one only in the two small real-panel effects noted above. The per-node variation that the law introduces is, at best, neutral.

Table 6: Attributing gains to the per-node law: every frozen configuration that runs a realization (law) against the identical pipeline with its α vector replaced at every step by its mean (mean twin) or by a random permutation.

Configuration	Panel	Law	Mean twin	Permuted	Law − mean [95% CI]	p
Rule + C on raw P, $\gamma = 4$, $K = 5$, $\beta = 0.5$	test	0.895	0.898	0.869	−0.003 [−0.005, −0.001]	2.3×10^{-7}
	real	0.889	0.882	0.864	+0.007 [+0.003, +0.011]	0.011
	$n = 900$	0.873	0.873	0.847	−0.000 [−0.004, +0.004]	0.002
Rule + A on raw P, $\gamma = 1$, $K = 1$, $\beta = 0.5$	test	0.851	0.853	0.775	−0.003 [−0.006, +0.001]	0.288
	real	0.862	0.855	0.773	+0.008 [+0.003, +0.012]	4.8×10^{-4}
	$n = 900$	0.824	0.823	0.752	+0.001 [−0.004, +0.006]	0.315
Rule + C on raw P, $\gamma = 1$, $K = 5$, $\beta = 0.5$ (rank)	test	0.775	0.898	0.860	−0.123 [−0.169, −0.080]	2.3×10^{-9}
	real	0.849	0.889	0.866	−0.039 [−0.068, −0.015]	0.813
	$n = 900$	0.772	0.879	0.841	−0.107 [−0.170, −0.052]	8.6×10^{-4}
C: EP, excess $\gamma = 16$, $T = 2$	test	0.766	0.763	0.747	+0.003 [+0.000, +0.005]	0.090
	real	0.823	0.825	0.792	−0.002 [−0.004, +0.000]	0.183
	$n = 900$	0.722	0.720	0.700	+0.001 [−0.002, +0.004]	0.722
B: EP, excess $\gamma = 16$, $T = 2$	test	0.762	0.763	0.680	−0.001 [−0.009, +0.005]	0.546
	real	0.820	0.825	0.698	−0.005 [−0.010, −0.000]	0.007
	$n = 900$	0.717	0.720	0.608	−0.003 [−0.010, +0.003]	0.798
A: raw P, excess $\gamma = 16$, $T = 2$	test	0.755	0.749	0.667	+0.006 [−0.010, +0.023]	0.895
	real	0.809	0.828	0.688	−0.019 [−0.031, −0.009]	0.001
	$n = 900$	0.728	0.733	0.598	−0.004 [−0.020, +0.011]	0.564
B: push, excess $\gamma = 16$, $T = 50$ (uniform pick)	test	0.623	0.733	0.589	−0.110 [−0.140, −0.081]	2.9×10^{-9}
	real	0.724	0.772	0.272	−0.047 [−0.069, −0.025]	2.6×10^{-5}
	$n = 900$	0.655	0.691	0.470	−0.036 [−0.062, −0.010]	0.002
C: raw P, $T = 3$ (Dirichlet pick)	test	0.634	0.888	0.762	−0.254 [−0.350, −0.166]	4.3×10^{-10}
	real	0.739	0.844	0.777	−0.105 [−0.168, −0.051]	4.2×10^{-4}
	$n = 900$	0.615	0.850	0.722	−0.235 [−0.364, −0.121]	0.004
Mode (b) as written: rule + B, $K = 1$, $\beta = 0.5$	test	0.728	0.731	0.724	−0.003 [−0.004, −0.002]	9.6×10^{-9}
	real	0.696	0.701	0.688	−0.005 [−0.007, −0.003]	3.3×10^{-10}
	$n = 900$	0.709	0.712	0.707	−0.003 [−0.004, −0.002]	3.8×10^{-6}

p: Wilcoxon test of the law-minus-twin difference.

6.6 Transfer to real networks and to larger graphs

Table 7 reports the frozen configurations on the real and scale panels, neither of which played any role in selection. On real networks the rule scores 0.840, the scalar oracle 0.897 and the per-node oracle 0.950; the per-node headroom over the scalar oracle is larger than on synthetic graphs (+0.053; +0.101 on Cora samples). The law does not capture it: AP($\eta = 1$) changes ρ by −0.015 and the law with the true y^* by −0.293. The dual-operator configurations transfer: Realization C on P gains +0.048, the constant-α control +0.060 [+0.047, +0.075], and the message-passing base polished by the scalar rule +0.060. At $n = 900$ (rule 0.832) the picture is the same: the constant-α control gains +0.054, Realization C on P +0.041, the anchored law changes ρ by −0.005, and the best standalone realization by −0.110. On the 40 instances where the per-node oracle runs at this size, it exceeds the scalar oracle by +0.033; there the oracle's time cap binds, so this headroom is understated.

6.7 Secondary metrics

The conclusions do not depend on the metric (Table 8). On the test panel, Kendall's τ rises from 0.737 for the rule to 0.793 for the constant-α control (scalar oracle 0.808), and the top-10% overlap from 0.583 to 0.688 for the Realization C polisher and 0.700 for its mean-field twin. On real networks the same configurations raise

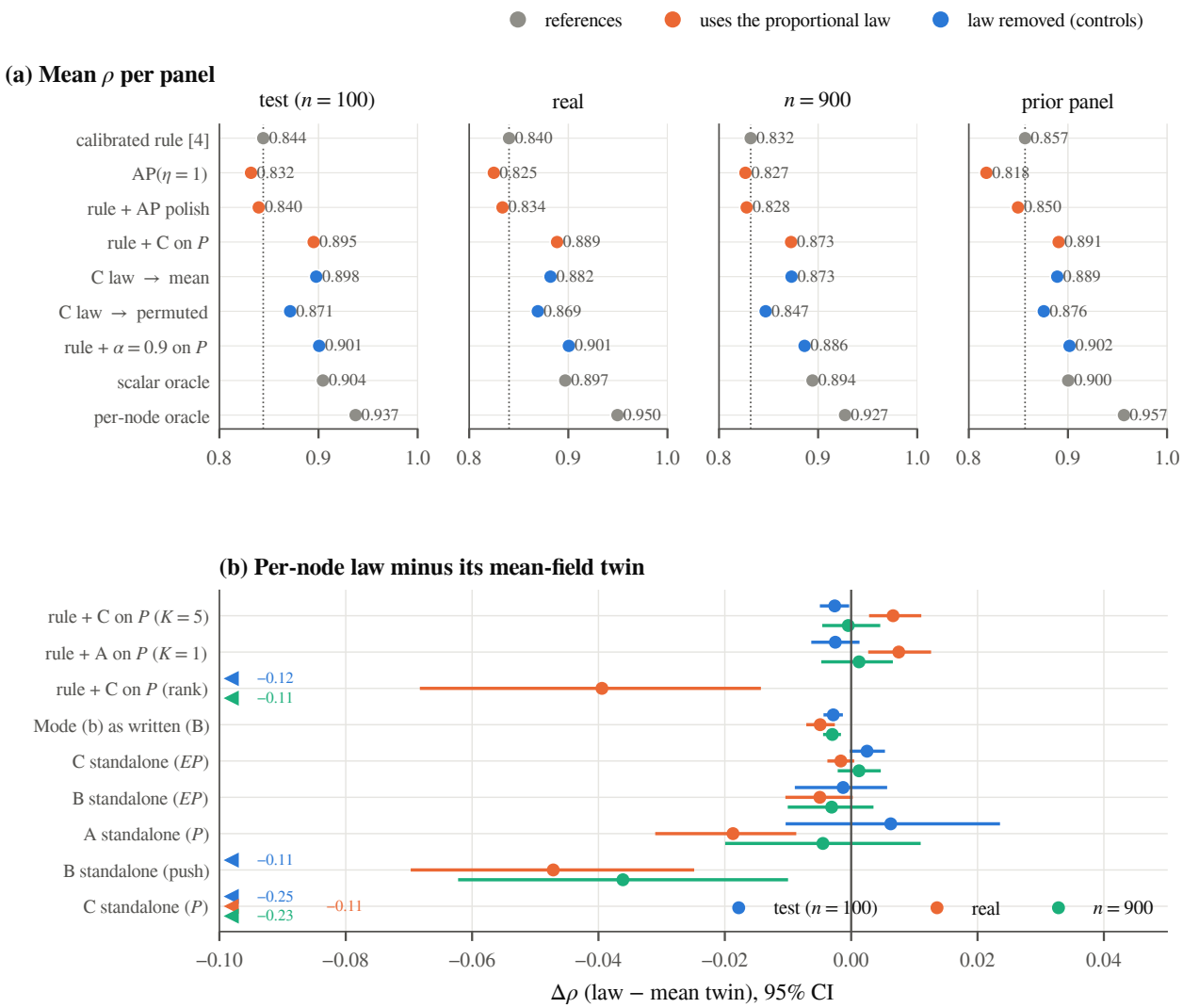


Figure 4: Mode (b) and its controls. (a) Mean ρ of references, configurations using the proportional law and matched controls on the four held-out panels (dotted line: calibrated rule). (b) Paired difference between every frozen configuration that runs a realization and its mean-field twin (same pipeline, α replaced by its per-step mean), with 95% bootstrap intervals; triangles mark differences below -0.10, whose values are printed.

NDCG@10% from 0.932 to 0.971 and 0.963. The anchored law and its polish are below the rule on all three metrics.

6.8 Why the dual-operator blend works

If the proportional law is not what improves Mode (b), the second operator must be. Table 9 isolates it with the configuration of Section 6.10: five sweeps of $y \leftarrow 0.1\,x + 0.9\,Py$ from scratch on the raw probability adjacency, with no calibrated base at all, already score 0.891 on the test panel (+0.046 over the rule), 0.864 on real networks and 0.858 at $n = 900$. The rank blend of the two improves on both members (0.904, 0.902 and 0.888), because their rank errors are only weakly correlated: the correlation of the per-vertex rank errors is 0.32 on the test panel, 0.17 on real networks and 0.33 at $n = 900$. The two operators err differently. The escape-scaled pull diffusion weights edges by $-\ln(1 - p_{ij})$ and normalises them per vertex, while P propagates raw probabilities and lets high-degree vertices accumulate mass, so the blend behaves like a two-member ensemble with decorrelated errors [27]. Realization C on P works for the same reason: its damping $S_i/(x_i + S_i)$ is close to one for most vertices once the base is embedded on the excess scale, so it is nearly a linear propagation on P, which is exactly what its mean-field twin shows.

Table 7: Transfer to real networks (88 instances) and to $n = 900$ (90 instances; the per-node oracle on 40 of them). p: Wilcoxon, Holm-corrected over the frozen configurations (†: uncorrected).

Method	$\bar{\rho}$ real	$\Delta\rho$ real [95% CI]	p	$\bar{\rho}$ n=900	$\Delta\rho$ n=900 [95% CI]	p
Calibrated scalar rule [4]	0.840	—	—	0.832	—	—
Scalar oracle	0.897	+0.057 [+0.041, +0.074]	$1.7\times10^{-13\dagger}$	0.894	+0.062 [+0.038, +0.090]	$8.4\times10^{-13\dagger}$
Per-node oracle	0.950	+0.109 [+0.091, +0.130]	$3.9\times10^{-16\dagger}$	0.927	+0.099 [+0.062, +0.142]	$6.4\times10^{-6\dagger}$
Law with true y^*: $\alpha_i = 1 - x_i/y_i^*$	0.547	−0.293 [−0.393, −0.202]	$7.2\times10^{-11\dagger}$	0.529	−0.303 [−0.432, −0.186]	$2.4\times10^{-7\dagger}$
AP($\eta = 1$), fixed point	0.825	−0.015 [−0.023, −0.009]	5.9×10^{-4}	0.827	−0.005 [−0.014, +0.005]	0.017
Rule + AP($\eta = 1$), $K = 1$, $\beta = 0.5$	0.834	−0.007 [−0.010, −0.004]	5.3×10^{-6}	0.828	−0.004 [−0.007, −0.002]	5.6×10^{-5}
Rule + AP($\eta = -0.25$), $K = 1$, $\beta = 0.25$	0.840	+0.000 [−0.000, +0.001]	1.000	0.832	−0.000 [−0.001, +0.000]	1.000
Mode (b) as written: rule + B, $K = 1$, $\beta = 0.5$	0.696	−0.144 [−0.191, −0.102]	1.3×10^{-4}	0.709	−0.123 [−0.169, −0.080]	1.9×10^{-5}
C: EP, excess $\gamma = 16$, $T = 2$	0.823	−0.017 [−0.051, +0.016]	1.000	0.722	−0.110 [−0.160, −0.062]	1.3×10^{-4}
MP + scalar rule, $K = 5$, $\beta = 0.75$ (rank)	0.900	+0.060 [+0.045, +0.077]	9.1×10^{-11}	0.877	+0.045 [+0.020, +0.072]	0.078
Rule + C on raw P, $\gamma = 4$, $K = 5$, $\beta = 0.5$	0.889	+0.048 [+0.033, +0.065]	1.2×10^{-7}	0.873	+0.041 [+0.019, +0.063]	0.054
Control: C law → per-step mean	0.882	+0.042 [+0.026, +0.058]	3.1×10^{-5}	0.873	+0.041 [+0.019, +0.064]	0.003
Control: constant $\alpha = 0.9$ on raw P (rank)	0.901	+0.060 [+0.047, +0.075]	9.9×10^{-13}	0.886	+0.054 [+0.038, +0.073]	2.7×10^{-6}

Table 8: Secondary metrics: Kendall's τ-b, top-10% overlap and NDCG@10% (relevance $= y_A^*$).

Method	τ test	top-10% test	NDCG test	τ real	top-10% real	NDCG real
Calibrated scalar rule [4]	0.737	0.583	0.963	0.702	0.464	0.932
Scalar oracle	0.808	0.658	0.981	0.776	0.570	0.959
Per-node oracle	0.852	0.708	0.987	0.869	0.668	0.976
AP($\eta = 1$), fixed point	0.717	0.559	0.953	0.682	0.451	0.918
Rule + AP($\eta = 1$), $K = 1$, $\beta = 0.5$	0.731	0.577	0.957	0.694	0.458	0.925
Rule + C on raw P, $\gamma = 4$, $K = 5$, $\beta = 0.5$	0.779	0.688	0.980	0.756	0.634	0.971
Control: C law → per-step mean	0.785	0.700	0.981	0.744	0.626	0.963
Control: constant $\alpha = 0.9$ on raw P (rank)	0.793	0.656	0.978	0.774	0.581	0.962

Table 9: Complementarity of the two operators: propagation $y \leftarrow 0.1x + 0.9Py$ for $K = 5$ from scratch (raw P) and the calibrated rule. The low correlation of their rank errors explains the gain of the blend.

Panel	ρ rule	ρ raw P	ρ rank blend	Rank-error correlation
test	0.844	0.891	0.904	0.32
real	0.840	0.864	0.902	0.17
$n = 900$	0.832	0.858	0.888	0.33

6.9 Cost, and the comparison with Monte Carlo

Table 10 and Figure 5 compare accuracy with single-process, end-to-end wall clock (operator construction, diffusion, embedding, sweeps and blending included). At $n = 100$ the calibrated rule takes 2.2 ms and the constant-α ensemble 2.7 ms, while a batched 200-world Monte Carlo estimate takes 1.7 ms and ranks better (0.906 against 0.901); 1000 worlds reach 0.945 in 6.2 ms. At this size, sampling is both cheaper and more accurate than every deterministic method in this paper. At $n = 900$ the order of costs reverses: the ensemble takes 4.6 ms against 12.5 ms for 200 worlds (2.7× faster) at equal accuracy (0.886 against 0.887).

To locate the crossover we timed the ensemble and Monte Carlo on 144 Erdős–Rényi, Barabási–Albert, Watts–Strogatz and grid instances with n from 100 to 2025 (range R_3, both priors; Table 11 and Figure 5c–d). The size at which the ensemble becomes faster than 200 worlds depends on the family: $n = 200$ on Erdős–Rényi graphs (Monte Carlo is faster at $n \leq 150$), $n = 200$ on Barabási–Albert graphs, $n = 576$ on grids, and between $n = 1300$ and $n = 2000$ on Watts–Strogatz graphs. Giving Monte Carlo the ensemble's time budget (the number of worlds that fit in the same time, interpolated in $\log k$) gives the fairer comparison. The median difference is −0.012 at $n \leq 150$ (the ensemble wins on 38% of the instances), −0.000 at $n = 400$–600, and +0.044 at $n \approx 2000$ (wins on 62%). The family matters as much as the size. The median difference is +0.022 on Erdős–Rényi graphs, −0.004 on Barabási–Albert graphs and +0.029 on grids, but −0.256 on Watts–Strogatz

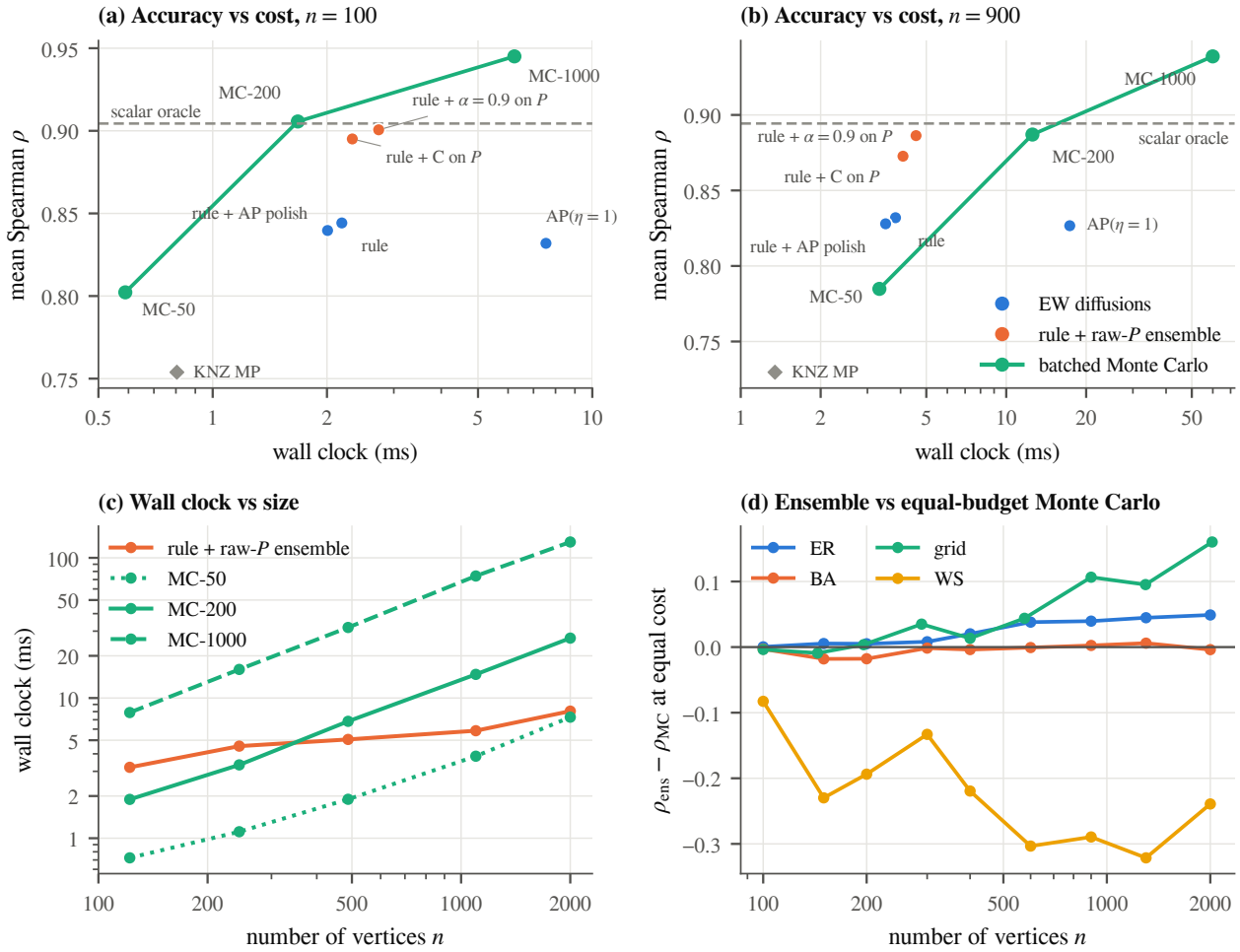


Figure 5: Accuracy, cost and Monte Carlo. (a, b) Mean ρ against single-process, end-to-end wall clock at $n = 100$ and $n = 900$ (log scale), with batched Monte Carlo at $k = 50$, 200 and 1000 worlds. (c) Wall clock against n for the dual-operator ensemble and batched Monte Carlo (size sweep, medians). (d) Difference in ρ between the ensemble and Monte Carlo given the ensemble's time budget, per family (medians).

graphs, where every deterministic method is far less accurate than sampling (median ρ 0.645 for the ensemble against 0.881 for 200 worlds). On grids at this range the calibrated rule alone is the more accurate deterministic method (0.980 against 0.962).

This corrects the cost argument of [4], which compared its heuristics with a sampler roughly an order of magnitude slower than a batched one. With batching, deterministic diffusions are worth their bias only on larger graphs and away from strongly clustered, near-critical structures; on the real panel even 50 worlds rank at 0.910 (200 worlds: 0.959), above every deterministic method we evaluated.

6.10 Exploratory: propagation on the raw adjacency alone

This subsection is exploratory and was not part of the frozen protocol. After the development and test results were known, we asked whether the second operator could be used without any calibrated base. We varied $a \in \{0.5, 0.9\}$, $K \in \{2, 5, 10\}$ and the blend weight $\beta \in \{0.5, 1\}$ for $y \leftarrow (1-a)x + aPy$ started from x — a truncated Katz-type score [25] with probability weights — either alone or rank-blended with the rule. The configuration $a = 0.9$, $K = 5$, $\beta = 0.5$ was the best on both panels. Since it was chosen after seeing them, only the real and scale panels can act as its hold-out: there it scores 0.902 and 0.888 (Table 9), in line with the frozen constant-α control (0.901 and 0.886). Raw-P propagation alone (0.864 and 0.858) is better than the rule on these panels but worse than either blend. Two cheap, structurally different propagations are better together than either is alone; whether the calibrated rule is the best partner for P is a question for a future pre-registered study.

Table 10: Accuracy (mean ρ over the whole panel) against single-process, end-to-end wall clock (median over one uniform-prior instance per family and probability range).

Method	$\bar{\rho}$ (n=100)	ms (n=100)	$\bar{\rho}$ (n=900)	ms (n=900)
KNZ message passing (2 it.)	0.754	0.80	0.730	1.35
Batched Monte Carlo, 50 worlds	0.802	0.59	0.785	3.33
AP($\eta = 1$), fixed point	0.832	7.55	0.827	17.35
Rule + anchored polish ($K = 1$)	0.840	2.01	0.828	3.51
Calibrated rule [4]	0.844	2.19	0.832	3.83
Rule + C on raw P ($K = 5$)	0.895	2.33	0.873	4.08
Rule + constant α on raw P ($K = 5$)	0.901	2.74	0.886	4.57
Batched Monte Carlo, 200 worlds	0.906	1.68	0.887	12.55
Batched Monte Carlo, 1000 worlds	0.945	6.23	0.939	59.94
Scalar oracle (needs ground truth)	0.904	—	0.894	—

Table 11: Size sweep (144 ER, BA, WS and grid instances, range R3, single process). "Ratio": MC-200 time over ensemble time; "equal-cost": Monte Carlo given the ensemble's time budget (interpolated in $\log k$). Medians per size class.

n	ms ensemble	ms MC-200	Ratio	Equal-cost k	ρ ensemble	ρ equal-cost MC	Difference	Wins
100–150	3.20	1.89	0.52	386	0.966	0.975	–0.012	38%
196–300	4.54	3.33	0.85	237	0.959	0.946	–0.009	41%
400–600	5.07	6.82	1.17	171	0.950	0.932	–0.000	47%
900–1300	5.84	14.75	2.65	76	0.952	0.899	+0.026	59%
2000–2025	8.06	26.75	3.53	60	0.940	0.879	+0.044	62%

7 Discussion

Why the principle fails. Four independent mechanisms, each proved in Section 4 and each visible in the data, explain the negative result. *Scale:* the coupled system holds at the target only if $Ty^* = y^* + x$, which no operator of spectral radius at most one allows (Remark 1, Proposition C); the realizations therefore try to reach a solution that the normalised operators do not have, and they collapse (Propositions A and B) or converge to a different law (Proposition D). *Stability:* where the active solution exists, it repels (Proposition C), so it cannot be found by iteration. *Sign:* the law raises the damping of well-connected vertices, which raises their scores only where $S_i > x_i$ (Proposition E); on normalised operators in sparse regimes this fails for most vertices, and the optimal per-node damping moves against the law (Section 6.3). *Prior:* under smooth heterogeneous priors the proportional ratio measures $1/x_i$ (Proposition 2′), and under seed priors it saturates. Anchoring and regularisation can make the law harmless — at $\eta = 0$ or $\tau = 1$, where it no longer is the law — but they cannot make it useful. Because the law with the *true* y^* fails as well, none of these failures is an artefact of estimation.

What per-node damping would need. Per-node damping does have out-of-sample headroom (+0.028 on synthetic graphs, +0.053 on real networks), but the oracle's damping is at best weakly aligned with the law, even where the sign condition holds (correlation at most 0.11). Proposition E indicates what a useful law must respect: at the operating point, raising α_k helps vertex k exactly when $S_k > x_k$, so any per-node rule has to depend on the sign of $S_k - x_k$ and not only on how much mass the vertex has accumulated. Designing and pre-registering such a rule is left for future work.

What Mode (b) contributes. The architecture of Mode (b) — a calibrated base diffusion, a few sweeps of a second propagation and a blend — is the part of the proposal that works, provided that the second propagation uses a *different operator*. A constant damping on the raw probability adjacency is the best second stage among the frozen configurations, costs K sparse matrix–vector products, transfers to real networks and to $n = 900$, and owes its gain to the weak correlation of the two operators' rank errors (Section 6.8). The proportional law can be placed in that stage without harm (Realization C), but its mean-field twin is equally good, and a constant is better.

Positioning against sampling. Deterministic ranking heuristics for uncertain graphs are often motivated by the cost of Monte Carlo sampling. With block-diagonal batching that cost is small: on graphs of a hundred vertices a 200-world estimate is cheaper and more accurate than every heuristic in this paper, and the deterministic ensemble is worth its bias only beyond a family-dependent size and away from clustered, near-critical structures such as Watts–Strogatz graphs. We recommend that evaluations of such heuristics include a batched sampler at matched wall clock.

Corrections to [4]. Two findings revise our earlier study. First, its calibration is prior-dependent: it reproduces under Dirichlet priors (96.2% of the scalar oracle) but recovers only 91.2% under the uniform prior, where a fixed constant beats it. Second, its cost comparison used a sampler much slower than a batched one; measured end to end, the deterministic advantage appears only on larger graphs. The motivating headroom of the principle (+0.062 overall, +0.341 on grids) re-measures at +0.028 and +0.018 out of sample.

8 Threats to validity

Ground-truth noise. With 2,000 worlds per sample, dense instances have unreliable ground-truth ranks; on dense Watts–Strogatz instances of the development panel the split-sample reference is only 0.29. All comparisons are paired, so this noise adds variance but no bias. *Oracle strength.* The per-node oracle is a coordinate ascent on a 14-value grid with at most three sweeps and a time cap; it is a lower bound in sample, and at $n = 900$ on lattices the cap binds, so the headroom there is understated, as it was in [4]. *Selection.* The development searches are large, which is why the winners were frozen before any held-out evaluation and why the held-out tests are Holm-corrected; the exploratory Section 6.10 is not covered by this protection. *Scope of the operators.* The law was tested on five operators and four scale conventions, standalone and as a polisher, anchored and unanchored, and with the true y^*. A different operator could behave differently, but Remark 1 and Proposition C constrain every operator: the target satisfies the system only on an expansive operator, on which the solution repels. *Reimplementation.* The calibrated rule uses the published constants of [4]; implementation details such as iteration counts and prior generators may differ from the original study, but the replication reproduces its Dirichlet-prior results. *Timing.* All timings are single-process Python and SciPy on one machine; ratios transfer better than absolute times, and compiled implementations of either side could move the crossovers. *Coverage.* The panels span nine synthetic families, five probability ranges, ten prior types and seven real networks with $n \leq 2025$; larger graphs and directed or correlated edge models remain untested.

9 Conclusion

The Proportional Relative Influence Principle couples a diffusion step with a per-vertex damping law, and its algebra is exact: the coupled system factorises as $(y_i - x_i)(y_i + x_i - S_i) = 0$. Its dynamics and its statistics are unfavourable, however. On the normalised operators used for ranking the realizations collapse to the prior, the self-consistent solution exists only on expansive operators and repels there, the law acts with the sign of $S_i - x_i$ rather than the reachability of the vertex, and heterogeneous priors turn the ratio into a measure of $1/x_i$. A pre-registered evaluation on 688 instances confirms every one of these mechanisms: no configuration that relies on the law beats the calibrated scalar rule of [4], not even with the true y^*. The hybrid architecture proposed alongside the law is nevertheless useful. Blending the calibrated diffusion with a few sweeps of propagation on the raw probability adjacency closes most of the gap to the scalar oracle at a cost of a few sparse products, and its gain comes from two operators that err differently. Finally, any claim of practical advantage for such heuristics must be measured against a batched Monte Carlo sampler, which is faster and more accurate on small graphs and is overtaken only beyond a family-dependent size.

Reproducibility statement

The archive accompanying this paper contains the library, the panel builder, the experiment runners, the analysis and control scripts, the per-instance results, the frozen configuration file with its SHA-256 digest, and the scripts that generate every number, table and figure of the paper (`make_paper_assets.py` and `paper_figures.py`). Every random draw is seeded deterministically, so rerunning a stage reproduces the stored results; rebuilding all macros, tables and figures from the stored results takes one command. The real-network source files are those distributed with [38]. The accompanying documentation gives the architecture of the code, complete proofs, a module-by-module walkthrough, the complexity of every component and a step-by-step reproduction guide.

References

[1] D. Kempe, J. Kleinberg, É. Tardos. Maximizing the spread of influence through a social network. In *Proc. 9th ACM SIGKDD*, 137–146, 2003.

[2] C.-Y. Lin, C.-H. Chin, H.-H. Wu, S.-H. Chen, C.-W. Ho, M.-T. Ko. Hubba: hub objects analyzer — a framework of interactome hubs identification for network biology. *Nucleic Acids Research*, 36:W438–W443, 2008.

[3] C.-H. Chin, S.-H. Chen, H.-H. Wu, C.-W. Ho, M.-T. Ko, C.-Y. Lin. cytoHubba: identifying hub objects and sub-networks from complex interactome. *BMC Systems Biology*, 8(Suppl 4):S11, 2014.

[4] I. Konstantellos. Damping from the escape probability: a free calibration rule for diffusion-based reachability ranking in uncertain graphs. Computer Engineering and Informatics Department, University of Patras, 2026. `https://zenodo.org/records/23211440`.

[5] B. Karrer, M. E. J. Newman, L. Zdeborová. Percolation on sparse networks. *Physical Review Letters*, 113:208702, 2014.

[6] J. S. Provan, M. O. Ball. The complexity of counting cuts and of computing the probability that a graph is connected. *SIAM Journal on Computing*, 12(4):777–788, 1983.

[7] M. Potamias, F. Bonchi, A. Gionis, G. Kollios. k-nearest neighbors in uncertain graphs. *Proceedings of the VLDB Endowment*, 3(1):997–1008, 2010.

[8] L. Page, S. Brin, R. Motwani, T. Winograd. The PageRank citation ranking: bringing order to the Web. Technical Report 1999-66, Stanford InfoLab, 1999.

[9] G. Jeh, J. Widom. Scaling personalized web search. In *Proc. 12th International World Wide Web Conference*, 271–279, 2003.

[10] T. H. Haveliwala. Topic-sensitive PageRank: a context-sensitive ranking algorithm for web search. *IEEE Transactions on Knowledge and Data Engineering*, 15(4):784–796, 2003.

[11] D. F. Gleich. PageRank beyond the Web. *SIAM Review*, 57(3):321–363, 2015.

[12] C. Borgs, M. Brautbar, J. Chayes, B. Lucier. Maximizing social influence in nearly optimal time. In *Proc. 25th ACM-SIAM Symposium on Discrete Algorithms*, 946–957, 2014.

[13] Y. Tang, X. Xiao, Y. Shi. Influence maximization: near-optimal time complexity meets practical efficiency. In *Proc. ACM SIGMOD*, 75–86, 2014.

[14] M. E. J. Newman. Message passing methods on complex networks. *Proceedings of the Royal Society A*, 479:20220774, 2023.

[15] P. Boldi, M. Santini, S. Vigna. PageRank as a function of the damping factor. In *Proc. 14th International World Wide Web Conference*, 557–566, 2005.

[16] M. Bressan, E. Peserico. Choose the damping, choose the ranking? *Journal of Discrete Algorithms*, 8(2):199–213, 2010.

[17] P. G. Constantine, D. F. Gleich. Random alpha PageRank. *Internet Mathematics*, 6(2):189–236, 2010.

[18] K. Avrachenkov, A. Kadavankandy, N. Litvak. Mean field analysis of personalized PageRank with implications for local graph clustering. *Journal of Statistical Physics*, 173:895–916, 2018.

[19] K. Avrachenkov, R. van der Hofstad, M. Sokol. Personalized PageRank with node-dependent restart. In *Algorithms and Models for the Web Graph (WAW 2014)*, Lecture Notes in Computer Science 8882, 23–33, Springer, 2014.

[20] X.-M. Wu, Z. Li, A. M.-C. So, J. Wright, S.-F. Chang. Learning with partially absorbing random walks. In *Advances in Neural Information Processing Systems 25*, 2012.

[21] J. Jung, W. Jin, H. Park, U. Kang. Supervised and extended restart in random walks for ranking and link prediction in networks. *PLOS ONE*, 14(3):e0213857, 2019.

[22] P. Li, E. Chien, O. Milenkovic. Optimizing generalized PageRank methods for seed-expansion community detection. In *Advances in Neural Information Processing Systems 32*, 2019.

[23] E. Chien, J. Peng, P. Li, O. Milenkovic. Adaptive universal generalized PageRank graph neural network. In *Proc. 9th International Conference on Learning Representations*, 2021.

[24] D. Berberidis, A. N. Nikolakopoulos, G. B. Giannakis. Adaptive diffusions for scalable learning over graphs. *IEEE Transactions on Signal Processing*, 67(5):1307–1321, 2019.

[25] L. Katz. A new status index derived from sociometric analysis. *Psychometrika*, 18(1):39–43, 1953.

[26] C. Dwork, R. Kumar, M. Naor, D. Sivakumar. Rank aggregation methods for the Web. In *Proc. 10th International World Wide Web Conference*, 613–622, 2001.

[27] A. Krogh, J. Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems 7*, 231–238, 1995.

[28] R. A. Horn, C. R. Johnson. *Matrix Analysis*, 2nd edition. Cambridge University Press, 2013.

[29] D. H. Lehmer. On the compounding of certain means. *Journal of Mathematical Analysis and Applications*, 36(1):183–200, 1971.

[30] P. Erdős, A. Rényi. On random graphs I. *Publicationes Mathematicae Debrecen*, 6:290–297, 1959.

[31] A.-L. Barabási, R. Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999.

[32] D. J. Watts, S. H. Strogatz. Collective dynamics of 'small-world' networks. *Nature*, 393:440–442, 1998.

[33] P. W. Holland, K. B. Laskey, S. Leinhardt. Stochastic blockmodels: first steps. *Social Networks*, 5(2):109–137, 1983.

[34] S. P. Borgatti, M. G. Everett. Models of core/periphery structures. *Social Networks*, 21(4):375–395, 2000.

[35] W. W. Zachary. An information flow model for conflict and fission in small groups. *Journal of Anthropological Research*, 33(4):452–473, 1977.

[36] D. E. Knuth. *The Stanford GraphBase: A Platform for Combinatorial Computing*. Addison-Wesley, 1993.

[37] A. Davis, B. B. Gardner, M. R. Gardner. *Deep South: A Social Anthropological Study of Caste and Class*. University of Chicago Press, 1941.

[38] O. Shchur, M. Mumme, A. Bojchevski, S. Günnemann. Pitfalls of graph neural network evaluation. Relational Representation Learning Workshop, NeurIPS 2018. arXiv:1811.05868.

[39] B. Efron, R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.

[40] F. Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

[41] S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[42] M. G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1–2):81–93, 1938.

[43] K. Järvelin, J. Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002.

[44] C. R. Harris et al. Array programming with NumPy. *Nature*, 585:357–362, 2020.

[45] P. Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.

[46] A. A. Hagberg, D. A. Schult, P. J. Swart. Exploring network structure, dynamics, and function using NetworkX. In *Proc. 7th Python in Science Conference*, 11–15, 2008.

A Additional tables

Table 12 lists the frozen configurations in the notation of the code, and Table 13 the branch fractions behind Figure 1a.

Table 12: The frozen configurations (`results/selected.json`).

Key	Type	Parameters
`A_best`	sa	`kind=A,op=P,scale=ex,gamma=16.0,T=2`
`B_best`	sa	`kind=B,op=EP,scale=ex,gamma=16.0,T=2`
`C_best`	sa	`kind=C,op=EP,scale=ex,gamma=16.0,T=2`
`C_dir1`	sa	`kind=C,op=P,scale=raw,gamma=1.0,T=3`
`B_unif`	sa	`kind=B,op=WT,scale=ex,gamma=16.0,T=50`
`AP_eta0.25`	ap	`eta=0.25,T=inf`
`AP_eta1`	ap	`eta=1.0,T=inf`
`PB_rule_best`	pb	`base=rule,eta=-0.25,mode=fix,T=1,beta=0.25,blend=s`
`PB_rule_nominal`	pb	`base=rule,eta=1.0,mode=fix,T=1,beta=0.5,blend=s`
`PB_rule_eta1`	pb	`base=rule,eta=1.0,mode=sc,T=inf,beta=0.25,blend=r`
`PB_mp2_best`	pb	`base=mp2,eta=0.25,mode=sc,T=5,beta=0.75,blend=r`
`PB_mp2_eta0`	pb	`base=mp2,eta=0.0,mode=sc,T=5,beta=0.75,blend=r`
`PB_mp2_eta1`	pb	`base=mp2,eta=1.0,mode=sc,T=5,beta=0.75,blend=r`
`PBX_rule_best`	pbx	`base=rule,kind=C,op=P,scale=raw,gamma=4.0,T=5,beta=0.5,blend=s`
`PBX_rule_unif`	pbx	`base=rule,kind=A,op=P,scale=raw,gamma=1.0,T=1,beta=0.5,blend=s`
`PBX_rule_dir1`	pbx	`base=rule,kind=C,op=P,scale=raw,gamma=1.0,T=5,beta=0.5,blend=r`
`PBX_rule_nominalB`	pbx	`base=rule,kind=B,op=EW,scale=ex,gamma=1.0,T=1,beta=0.5,blend=s`
`CTRL_Cmean`	ctrl	`law=Cmean,gamma=4.0,T=5,beta=0.5,blend=s`
`CTRL_Cperm`	ctrl	`law=Cperm,gamma=4.0,T=5,beta=0.5,blend=s`
`CTRL_lin0.5`	ctrl	`law=lin0.5,gamma=4.0,T=5,beta=0.5,blend=s`
`CTRL_lin0.9`	ctrl	`law=lin0.9,gamma=1.0,T=5,beta=0.5,blend=r`

Table 13: Fractions of vertices on the trivial branch ($y_i = x_i$) and the active branch ($S_i > 2x_i$) over the iterations (development panel).

Configuration	Prior	trivial t=1	trivial t=5	trivial t=20	trivial t=200	active t=200
A, EW, raw	uniform	1.00	1.00	1.00	1.00	0.00
	Dir(1)	0.61	0.70	0.70	0.70	0.30
B, EW, raw	uniform	0.00	0.77	1.00	1.00	0.00
	Dir(1)	0.00	0.51	0.57	0.69	0.30
C, EW, raw	uniform	0.00	0.00	0.00	0.00	0.00
	Dir(1)	0.00	0.00	0.00	0.00	0.30
A, EW, L_1	uniform	1.00	1.00	1.00	1.00	0.00
	Dir(1)	0.00	0.00	0.00	0.00	0.25
B, EW, L_1	uniform	0.00	0.22	0.64	1.00	0.00
	Dir(1)	0.00	0.00	0.00	0.00	0.25
A, EW, excess $\gamma = 16$	uniform	0.07	0.13	0.21	0.38	0.62
	Dir(1)	0.35	0.53	0.57	0.58	0.42
C, P, raw	uniform	0.00	0.00	0.00	0.00	0.58
	Dir(1)	0.00	0.00	0.00	0.00	0.69